IP5X00 Console API

Revision 2.17

April 8, 2025

Revision History

Version	Date	Editor	Notes
1.0	May 8, 2016	Sid	Initial
1.1	Jun 22, 2016	Sid	Update EDID Video States etc.
1.2	Jun 26, 2017	Sid	Update audio selection for TX.
1.3	Jun 30, 2017	Sid	Update BMP & JPG preview API
1.4	Mar 12, 2018	Sid	Update API including Source Selection V2 API, Sending IR, Analog Audio Volume Control, Video Status, Sink EDID
1.5	Dec 19, 2018	Sid	Update "Bit Rate Setting", "Video Status", "VW v2", "Turn Off/On Video Output", including Appendix A.
1.6	Jan 17, 2019	Sid	Add KMoIP Roaming, including Appendix B.
1.7	Jan 23, 2019	Sid	Update API of 'cec_send'
1.8	Oct 30, 2019	Sid	Add 'HDMI Mute Control' Modify 'Audio Setting', add new of it. Add 'OSD Display Settings' Modify 'Output Resolutions Settings', add genlock scaling. Add 'USB Hotkey settings'
1.9	Nov 19, 2019	Sid	Add 'Video input Selection'
1.10	Sep 17, 2020	Sid	Correct 'soip2' with hex mode support
1.11	Nov 9, 2020	Sid	Add e_stop_link
1.12	Oct 10, 2021	Sid	Add ‘usb_force_fullspeed’ support
1.13	Apr 18, 2022	Sid	Add IP5100 support
1.14	June 1, 2022	Sid	Add gbstatus report. Add Audio Return Path support
1.15	Sep 16, 2022	Sid	Add e_find_me API
2.0	Feb 23, 2023	Sid	Change the document to new organization
2.1	Mar 9, 2023	Sid	Add "e_a_direction" API
2.2	Mar 15, 2023	Sid	Add "e_v_switch_in" API Add "e_v_switch_mode" API

			Add "e_v_switch_pri" API
2.3	April 17, 2023	Sid	Correct ' reset_to_factory.sh ' to 'reset_to_default.sh '
2.4	April 20, 2023	Sid	Add 'Audio over IP' about dante related astparam for ethernet port and IP addresses
2.5	July 27, 2023	Hornet	Add 'e_usbc_charger' about USB-C charger
2.6	July 31,2023	Sid	Correct the free routing related document.
2.7	Aug 18, 2023	Sid	Correct 'multicast_on' default value.
2.8	Aug 21, 2023	Sid	Add astparam 'ss_snapshot_option' and ' ss_stream_option' for preview(sub-stream) configuration.
2.9	Dec 29, 2023	Sid	Update 'e_reconnect' Add 'e_discovery' with LLDP support, Add event 'e_edid_fast_switch_mode', 'e_osd_menu_font_size', 'e_osd_menu_hot_key', 'e_osd_menu_position', 'e_osd_menu_time_out_sec', and related 'astapram'
2.10	April 21, 2024	Sid	Add 'e_aes67_cli_start' and 'e_aes67_cli_stopt' for AES67 subscription. Add 'Sink Power Control' section
2.11	April 24, 2024	Sid	Add 'dante_op_mode' for Dante AV-A operation mode configuration.
2.12	July 24, 2024	Sid	> Update "a_addon" for support aes67 > Add "addon_port" for support assign separate port to audio addon > Add "http://ip/device/log" for log export > Add 'e_ntp_on', 'e_ntp_off', 'enable_ntp_cli', 'ntp_srv_addr', 'e_set_time_zone', 'time_zone' for NTP support > Update 'vw_stretch_type' for new Fit Video Wall Mode > Add 'usb_board_type', 'e_switch_board_type' for IPT5100U change board working mode

2.13	Sep 18, 2024	Sid	> Add 'e_set_aesmode'
2.14	Nov 6, 2024	Sid	> Add 'addon_ip_mode', 'addon_ipaddr', 'addon_netmask' for configurating Dante/AES67 to 2nd port on specific model.
2.15	Nov 15, 2024	Sid	> Correct the force output timing sysfs API.
2.16	Jan 13, 2025	Sid	> Correct 'e_osd_menu_hot_key', 'e_osd_menu_time_out_sec' related 'astapram'
2.17	April 8, 2025	Sid	> Add below events and related astparam: - e_lan_mode - e_addon_ethernet - e_aes67_session - e_web_passwd_change - e_usb_b_ethernet - e_usb_c_ethernet - e_enable_usb_b_network - e_enable_usb_c_network

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Overview

The "Console APIs" interface is so called "Command Line Interface" (CLI). This document describes the commands which can be used under IP5000 & IP5100 series product's console interface. IP5000 and IP5100 uses Linux OS and the console is derived by Bash shell. Console can be accessed through:

- Telnet port 24: There is built-in telnet server to connect to IP5000 & IP5100 devices.
- SSH port 22: There is built-in SSH server to connect to IP5000 & IP5100 devices.

Through console APIs, developers can get more control on IP5000 & IP5100 firmware and extend product's features and capability.

Supported Products

- IPE5000 / IPD5000
- IPE5000W
- IPE5000DNT
- IPE5100 / IPD5100
- IPE5101
- IPE5100P / IPD5100P
- IPE5102 / IPD5102
- IPT5100U

Preface

How to Access the Console

There are several ways to access the console:

1. **Use telnet**: FW has built-in 'telnetd' server built in. Developer can use any telnet client to connect to any IP5000 and IP5100 device. Or write their own program using telnet protocol to connect to any IP5000 and IP5100 device through IP network.
   NOTE: Default use telnet port 24 instead of 23. Use "root" to login. No password is required for login.
2. **Use SSH**: [FW >= 2.5] FW has built-in SSH server. Developer can use any SSH client to connect to any IP5000 and IP5100 board.
   NOTE: use “root” to login. No password is required for login.

How to get IP5X00 IP address

FW default runs "DHCP" as an IP address assignment method, if there is no DHCP server available, device will automatically be fallback to a link local IP address which in 169.254.0.0/16 subnet. The IP address can be also resolved by its hostname using mDNS (Bonjour) and SSDP protocol.
Here are the methods to resolve the IP address of IP5X00 devices.

1. Decoder connected display: The decoder connected display will display the decoder's IP address and connected encoder's IP address.
2. mDNS (Bonjour) discovery
3. SSDP discovery
4. IP5X00 discovery with C language and python3 demo we provided. (Please get from the technical support)

Command Category

Different usage scenario is achieved by different types of commands.

"Console APIs" section describes all those commands by usage type.

However, by implementation type, Commands can be classified into following types:

- Link manager control:
    - ‘e’ command. Send event to link manager.

    - ‘lmparam’ command. Query link manager parameters.
- Flash access control:
    - ‘astparam’ command (alias as 'gbparam' too). Used to read/write parameters into flash. See "List of astparam" for details.
- Driver control:
    - ‘sysfs’ driver files. Drivers use Linux ‘sysfs’ filesystem to export proprietary control interface to user.
- Bash shell commands:
    - echo, cat, reboot…. Normally used to access ‘sysfs’.
- Special console commands:
    - Some bash shell scripts or other binary commands.

Command Syntax Conventions

Following table describes the syntax used with the commands in this document.

<table>
<tr><th>Convention</th><th>Description</th></tr>
<tr><td>boldface</td><td>Commands and keywords.</td></tr>
<tr><td>italic</td><td>Command input that is supplied by you.</td></tr>
<tr><td>[x]</td><td>Keywords or parameters that appear within square brackets are optional.</td></tr>
<tr><td>{x}</td><td>Keywords or parameters within braces must be entered.</td></tr>
<tr><td>x|y</td><td>Keywords or parameters separated by a vertical bar require you to choose one option.</td></tr>
<tr><td>x||y</td><td>Keywords or parameters separated by a double vertical bar allow you to choose any or all of the options.</td></tr>
<tr><td>_</td><td>Just a space, not the underline.</td></tr>
<tr><td></td><td></td></tr>
<tr><td></td><td></td></tr>
<tr><td></td><td></td></tr>
</table>

Note:

- All commands and parameters are case sensitive.
- "*/ #* " in command examples means command prompt. It is not part of command.

Console APIs

Link Manager Control

Send Event to Link Manager

This is the most used command to control the system. This command has two syntax formats. 'e' is just a shortcut of 'ast_send_event -1'. They are basically the same. The detail usage of events will be described in the reset of this document. And a list of available events are listed in section "List of Link Manager Events".

Syntax

e {*event*[**::***arg1*][**::***arg2*]...}

Parameters

> event: A case sensitive string to represent a event. Usually starts with 'e_'.

- Ex:

```
e e_reconnect
```

> arg: If an event need some arguments, it is usually attached after event string with '::' token. Some events uses '_' to separate arguments.

```
e e_reconnect::123456789012::vua
```

Get Link Manager Internal Parameters

lmparam is usually used to query some of link manager's internal parameters. This command also provides a method to set the value of specified parameter. However, it is rarely useful and may confuse link manager script results in unexpected problem.

Syntax

lmparam g {*key*}

lmparam dump

Parameters

- g: Get value of specified key. When 'g', no [*value*] is needed.
- key: **"**List of lmparam**"** has a list of useful lmparam keys.
- dump: Dump all lmparam key/value.

NOTES

- Link manager internal parameters is implementation dependent. Which means their definition may changes when FW changes. Please avoid using it, especially parameters that are not mentioned in this document.

Examples

Get current routed video encoder MAC ID

```
/ # lmparam g CH_SELECT_V
123456789012/ #
```

Flash Access Control

The most fundamental and powerful way to access astparam is through the "astparam" Console API command. Most of alternative approaches are just a wrap of "astparam" command. Please reference.

Syntax

astparam {*OPTION*} {*KEY*} [*VALUE*]

astparam dump

Parameters

- *OPTION:*
  - g: read from RW partition cache file, {*KEY*} is required when {*OPTION*} is 'g'
  - s: write to RW partition cache file (**not save to flash ROM yet**). {*KEY*} and {*VALUE*} are required. If {*VALUE*} is empty the {*KEY*} will be removed.
  - flush: clear all settings in RW partition cache file. "astparam save" is needed to clear all settings in flash ROM.
  - dump: dump all parameters in RW partition cache file.
  - save: **save all parameters in RW partition cache file into flash ROM**.
- *KEY*: Section "List of astparam" has a list of useful lmparam keys.
- *VALUE*: The value for [*KEY*]

Response

OPTION = 'g': please reference Example-2 and Example-3 below

OPTION = 'dump': please reference Example-4 below

NOTES

Without '**astparam save**', after unit reboot, all *KEY*/*VALUE* will lost.

Examples

Example-1: Set IP addresses

```
/ # astparam s ip_mode static
/ # astparam s ipaddr 192.168.1.2
/ # astparam s netmask 255.255.255.0
/ # astparam s gatewayip 192.168.1.1
/ # astparam save
/ #
```

Example-2: Get IP mode set

```
/ # astparam g ip_mode
static/ #
```

Example-3: Get IP mode set

```
/ # astparam g ip_mode
static/ #
```

Example-4: Dump all *KEY*s *VALUE*s

```
/ # astparam dump
CRC = 0xED9C8FA6
ch_select_v=123456789012
ch_select_u=123456789012
ch_select_a=123456789012
ch_select_s=123456789012
ch_select_r=123456789012
ch_select_p=123456789012
ch_select_c=123456789012
/ #
```

Driver Control

‘sysfs’ driver files. Drivers use Linux ‘sysfs’ filesystem to export proprietary control interface to user. Please find all available driver control in section "List of Driver sysfs"

Others

Other command could be busybox commands, or some scripts, please refer section "List of other Commands"

Appendix A. List of astparam

Generic:

Key	Description	Value (bold is default)	Encoder/Decoder
account_info	Used to record account info. This will be used to configure ssh server and telnet server's root password. Format: {*USER*},{*PASSWORD*},{*ROLE*}[:{*USER*},{*PASSWORD*},{*ROLE*}{:...}] > *USER*: username > *PASSWORD*: Password encrypted by DES. > *ROLE*: ○ admin: root level ○ user: not support yet. **NOTE:** > this astparam can ONLY be set by e_set_account command. Please reference to 'List of Link Manager Events'. > admin's username MUST be 'root' > Default has only 'root' user Example astparam value: root,RYfwLn00.RcBA,admin:user,HO6bloKUqFZ5Y,user	See description	Encoder/Decoder
bonjour_enable	Enable 'Bonjour' service	y: Enable 'Bonjour' **n: Disable 'Bonjour'**	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
en_log	‘y’ show debug message on debug console.	y: Enable debug log **n**: Disable debug log	Encoder/Decoder
enable_ntp_cli	Enable NTP client.	y: Enable **n**: Disable	Encoder/Decoder
free_routing	Enable ‘free_routing’ feature. See ‘mcip_srv_map’ for details. Disable it can save multicast groups requested. Regular please keep it as 'y'.	**y**: Enable n: Disable	Encoder/Decoder
lldp_enable	Enable 'LLDP' service	y: Enable 'LLDP' **n: Disable 'LLDP'**	Encoder/Decoder
lm_link_off_timeout	The timeout in seconds for link to encoder.	0 ~ **8**: 8 seconds	Decoder
multicast_on	Configure casting mode	**y**: Multicast mode n: Unicast mode	Encoder/Decoder
ntp_srv_addr	NTP server IP address or URL	**0.0.0.0**	Encoder/Decoder
service_https_enable	Enable 'https'.	**n**: Disable. just http y: Enable https and redirect all http to https	Encoder/Decoder
service_ssh_enable	Enable 'ssh' server.	**y**: Enable ssh n: Disable ssh	Encoder/Decoder
service_telnet_enable	Enable 'telnet' server.	**y**: Enable ssh n: Disable ssh	Encoder/Decoder
ssdp_enable	Enable 'SSDP' service	y: Enable 'SSDP' **n: Disable 'SSDP'**	Encoder/Decoder
telnetd_param	‘-p 24’: use IP port 24. Please reference to busybox manual for details. Ex:	**-p 24**	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
	'-p 23 -l /bin/sh': Use port 23 without login. '--help': Disable telnetd. It is a trick to stop telnetd from loading.		
time_zone	Setup time zone	**Etc/UTC**	Decoder
ui_default_res	Format: {*WIDTH*}x{*HEIGHT*}@{*FPS*}[,{*FORMAT*}] > *WIDTH*: resolution width > *HEIGHT*: resolution height > *FPS*: frames per second > *FORMAT*: 0 - DVI, 1 - HDMI, optional with default DVI For example: 640x480@60 => 408P, DVI 640x480@60,1 => 480P, HDMI **Note**: system reboot is required for ui_default_res to take effect.	**1280x720@60,1**	Decoder
ui_show_text	'y' - display diagnostic information on the bottom of GUI. 'n' - hide diagnostic information.	**y:** Display n: Hide	Decoder

Network Setting:

Key	Description	Value (bold is default)	Encoder/Decoder
ch_select_a	Audio channel selection		Decoder
ch_select_c	CEC channel selection (IP5100 only) (Not support now)		Decoder
ch_select_i	ARP channel selection (IP5100 only)		Encoder
ch_select_r	IR channel selection		Decoder
ch_select_s	Serial channel selection		Decoder
ch_select_u	USB channel selection		Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
ch_select_v	Video channel selection		Decoder
gatewayip	Gateway IP address		Encoder/Decoder
en_usb_b_network	Enable USB-B Ethernet	**y** n	Encoder (IPE5102)
en_usb_c_network	Enable USB-C Ethernet	**y** n	Encoder (IPE5102)
ipaddr	IP address		Encoder/Decoder
ip_mode	IP mode	static: **dhcp:** autoip:	Encoder/Decoder
jumbo_mtu	Ethernet MTU size in bytes. Only listed value can be used. Both encoder and decoder must config to the same value. **Note: IPX5100 Default value is 1500 but please enable Ethernet switch's jumbo frame setting when Video IP doesn't work.**	**1500** (IPX5100) 1600 8000 (IPX5000)	Encoder/Decoder
lan_mode	Set Dual RJ45 LAN ports product, the LAN ports working mode	**Switched** bridged independent	Encoder/Decoder
multicast_ip_prefix	It will be the 'aaa' part of multicast IP, aaa.bbb.ccc.ddd. Detail, please reference "IP5X00 Multicast Address & Port List" document.	**234**	Encoder/Decoder
mcip_srv_map	A list of number separated by ':'. Used to configure channel to multicast IP mapping. It is used to map 'free_routing' multicast IP'. When 'free_routing' is 'y', the default value of 'mcip_srv_map' is '2:3:4:5:8:7:9:6'. Otherwise, will be '2:2:2:2:2:2:2:2'.	**2:3:4:5:8:7:9:6** : if 'free_routing' is 'y' **2:2:2:2:2:2:2:2** : when 'free_routing' is 'n'	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
	This list maps to different services: **video:audio:IR:serial:pushbutton:USB:CEC:ARP** Detail, please reference "IP5X00 Multicast Address & Port List" document. NOTE: > Both encoder and decoder must use the same setting to work. > Pushbutton is not supported by hardware. > CEC is not available now		
multicast_leave_force	Auto send IGMP leave packet when received multicast packets that are not interested in. Specify a value higher than 0 will enable this feature. However, DO NOT use a value that is too small. The smaller the value the higher load Ethernet switch may impact. Just use default value should be good to go. The specified time interval is used to specify the minimum interval of sending IGMP leave packet.	**8000**: Enable with 8000 ms interval 0: Disable	Encoder/Decoder
netmask	Netmask address		Encoder/Decoder
usb_b_ethernet	Set USB-B Ethernet LAN port	**lan1** lan2	Encoder (IPE5102)
usb_c_ethernet	Set USB-C Ethernet LAN port	**lan1** lan2	Encoder (IPE5102)

Video over IP:

Key	Description	Value (bold is default)	Encoder/Decoder
ast_video_quality_mode		**-1**: Video mode 0: Graphic mode (Best) 1: 2: 3: 4: 5: (Worst)	Encoder
edid_fast_switch_mode	Set EDID mode for fast switching, reduce Extended Color, HDR, 4:2:0 capability from any imported EDID.	on **Off**	Encoder
edid_use	**Unicast Mode**: All decoder's 'edid_use' will **ALWAYS** be 'primary'. **Multicast Mode**: All encoder and decoder's 'edid_use' default is 'secondary'. The primary EDID will be used to the EEPROM on connected encoder. Note: **In most of cases, leave 'edid_use' as default will be fine.**	primary: **secondary**:	Decoder
video_genlock_scaling	Enable genlock scaling mode, which always scaling timing width higher than 1080 to 1080P, but keep frame rate as input stream.	y: Enable **n**: Disable	Decoder
no_video	Disable Video over IP	y: Disable VideoIP **n**: Enable VideoIP	Encoder/Decoder
preview_enable	Enable/Disable preview stream/snapshot	**y**: Enable n: Disable	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
profile	Revised the naming to: auto, 10M, 50M, 100M, 150M, 200M.	**auto**: 800Mbps for IP5100 10M: 50M: 100M: 150M: 200M:	Encoder
ss_snapshot_option	The snapshot configuration. It is a string in following format: WIDTH,HEIGHT,QUALITY,AS WIDTH/HEIGHT is in pixel unit. QUALITY range: 10~100 (in step 10), higher setting means better image quality. AS: aspect ratio configuration. - 0: extend to what [WIDTH] and [HEIGHT] configured - 1: keep original aspect ratio and place in the center of output (letterboxing or pillar boxing)	**1024,576,60,0**	Encoder
ss_stream_option	The sub-stream configuration. It is a string in following format: WIDTH,HEIGHT,FPS,BW,AS,MINQ WIDTH/HEIGHT is in pixel unit. FPS: frame rate. Unit: frame per second BW is in Kbps MINQ: minimum quality number. range: 10, 20, …, 90, 100 AS: aspect ratio configuration. - 0: extend to what WIDTH and HEIGHT configured	**960,540,15,8000,0,60**	Encoder

Key	Description	Value (bold is default)	Encoder/Decoder
	- 1: keep original aspect ratio and place in the center of output (letterboxing or pillar boxing)		
v_frame_rate	Used to control frame rate of video encode. Default set to ‘0’ means ‘no limitation’ use best effort. Set to other value will limit the maximum video encode frame rate. Valid range: 0,1,2,...,60. Frame rate = (v_frame_rate / 60) * input refresh rate For example, set to ‘30’ under 1080p60Hz: Frame rate = (30/60) * 60 = 30 FPS	**0: Best effort**	Encoder
v_gen_lock_cfg	**Don’t touch this value unless you are told to.** The value is 32bits hex value: gen_lock_cfg[31:24]: location. the location user prefers to lock at (in percentage) gen_lock_cfg[22:8] ppm limit. acceptable PPM variation of monitor gen_lock_cfg[7:4] factor. number of switches between encoder and decoder gen_lock_cfg[3]: ppm limit 4k patch: use 10% ppm limit for 4K resolution. gen_lock_cfg[2]: ppm strict mode: limit PPM even if in JUMP state. gen_lock_cfg[0]: enable:	**55028aa9**	Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
	enable/disable gen-lock Default value is ‘55028aa9’. Set to ‘0’ to disable video gen lock.		
v_hdmi_hdr_mode	Set decoder HDMI output hdr drop.	**0**: HDR passthrough 1: Force HDR off	Decoder
v_output_timing_convert	Force specific video output timing. Examples: 00000000: Pass-Through (Strict Mode) 10000000: Pass-Through (Strict Mode) 82000000: Base on EDID 80000061: Ultra HD 2160p60 80000060: Ultra HD 2160p50 8000005F: Ultra HD 2160p30 8000005E: Ultra HD 2160p25 8000005D: Ultra HD 2160p24 80000010: Full HD 1080p60 8000001F: Full HD 1080p50 80000022: Full HD 1080p30 80000021: Full HD 1080p25 80000020: Full HD 1080p24 80000004: HD 720p60 80000013: HD 720p50	**0**: Pass-Through	Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
	8000003E: HD 720p30 8000003D: HD 720p25 8000003C: HD 720p24 81004048: WXGA 1366x768@60 81004021: WXGA+ 1440x900@60 81004032: WUXGA 1920x1200@60 8100401D: SXGA+ 1400x1050@60		
v_pwr_save_timeout	Wait for a v_pwr_save_timeout ms before power save starts. When decoder is streaming video and video source lost, decoder will wait for `v_src_unavailable_timeout' + 'v_pwr_save_timeout' ms time before firing 'e_video_enter_pwr_save' event (and enter power save). When decoder starts (not streaming video), it will wait for ''v_pwr_save_timeout' ms time before firing 'e_video_enter_pwr_save' event (and enter power save).	**8000:** 8 seconds	Decoder
v_src_unavailable_timeout	Wait for a v_src_unavailable_timeout ms before entering power save. Set to '-1' means never timeout.	**10000**: 10 seconds -1: never timeout	Decoder
v_switch_in	The switched input for multi-input encoder when it is manually mode.	**hdmi1** hdmi2 usb-c	Switch Encoder
v_switch_mode	Switch mode for multi-input encoder.	**auto** manual priority	Switch Encoder
v_switch_pri	Switch priority when switch mode is priority mode.	**hdmi1:hdmi2:usb-c**	Switch Encoder
v_turn_off_screen_on_pwr_save	Turn off video output sync after connected encoder stops streaming video to decoder.	y: Turn off	Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
		n: Do not turn off. Will switch back to GUI screen	

Video Wall:

Key	Description	Value (bold is default)	Encoder/Decoder
en_video_wall	Enable/Disable video wall feature.	**y**: enable. n: disable.	Decoder
vw_rotate	Clockwise rotate and Flip	**0**: No rotate 1: vertical flip 2: horizontal flip 3: 180 degrees 5: 90 degrees (IPD5100) 6: 270 degrees	Decoder
vw_stretch_type	Video wall stretching type: 1: Stretch out (keep stream ratio) 2: Fit in (stretch stream ratio, for fitting display)	1: Crop to fill (Stretch out) **2**: Stretch (Fill in) 3: Crop to fill (Stretch out) 4: Fit (Black board fit)	Decoder
vw_ver	Specify video wall control APIs version: 1: Default. Specify monitor layout and auto calculate coordinate. 2: Manually specify coordinate. Easier for mosaic style video wall configuration. Please see “Appendix G” for details.	**1**: Default 2: v2. Specify coordinate	Decoder

HDCP:

Key	Description	Value (bold is default)	Encoder/Decoder
hdcp_always_on	Always enable HDCP 1.4	y: enable **n**: disable	Encoder/Decoder
hdcp_always_on_22	Always enable HDCP 2.2	y: enable **n**: disable	Encoder/Decoder
hdcp_enable	Set encoder HDCP enable or disable. Use 'e e_adj_hdcp' for taking effect immediately	**y**: enable n: disable	Encoder

USB over IP:

Key	Description	Value (bold is default)	Encoder/Decoder
no_usb	Enable/Disable USB feature	y: **n**:	Encoder/Decoder
share_usb	Default value: (when ‘share_usb_auto_mode’ is ‘n’) Multicast mode: ‘share_usb = y’. Unicast mode: ‘share_usb = n’.	**y**: n:	Encoder/Decoder
share_usb_auto_mode	When ‘share_usb_auto_mode == y’, ‘share_usb’ setting will automatically change depending on the setting of casting mode (‘multicast_on’).	y: **n**:	Encoder/Decoder
share_usb_on_first_peer	Applied to ‘share_usb == y’ mode. When value is ‘y’, the first connected decoder will automatically request USBoIP function. By enabling this feature, USBoIP will automatically enabled when there is only one decoder connected to encoder under multicast mode.	**y**: auto connect first peer n:	Encoder

Key	Description	Value (bold is default)	Encoder/Decoder
usb_board_type	Set the IPT5100U board working on host (TX or encoder) mode or client (RX or decoder) mode NOTE:This is only available for IPT5100U	**per GPIO (DIP switch)** host client	IPT5100U
usb_force_fullspeed	For decoder USB working on full speed mode. (IPD5000 Only)	**n**: Disable y: Enable	Decoder
usb_hid_urb_interval	Applies to both KMoIP and USBoIP. Set to \`35\` to resolve some USB HID long latency issue. Set to \`7\` to resolve ELO USB touch panel issue.	**0:** ASAP 35: 35ms	Encoder
usb_quirk	32bits bitmap options for resolving some vhub/USB HC compatibility issues. 0x00000001: [Encoder] patch SSPLIT packet type to BULK type. Ex: Android 4.4.2 + CVTouch 0x00010000: [Decoder] extend maximum packet size to the maximum value defined in specification. Please try this option if you encountered decoder device error messages like \`usb 3-2: device descriptor read/64, error -110\`. Ex: logitech G403 mouse. 0x00020000: [Decoder] No clear halt. Some USB devices (ex: HSL KM switch) don't like \`clear halt\` command and will behave unexpected. And it may stop decoder USB from working until decoder reboot. Enable this bit to resolve this issue. 0x00040000: [Decoder] Don't reset USB twice. Ex: Elo Touch Solutions E756274 ET2402L.	**0:**	Encoder/Decoder

USB over IP Exporting Policy:

Key	Description	Value (bold is default)	Encoder/Decoder
usb_default_policy	The default/global exporting policy used when a specific policy is not available.	**auto_export**	Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
		no_auto_export	
usb_conflict_policy	Policy used when a device has different interfaces with conflict exporting policy.	auto_export **no_auto_export**	Decoder
usb_disable_classes	Policy setup to disable per USB classes. There is no default value. Each class should be separated by a space. For example, **usb_disable_classes=audio video printer**.	per_interface audio comm hid physical still_image printer mass_storage cdc_data cscid content_sec video personal_healthcare audio_video diagnostic wireless_controller misc app_spec vendor_spec billboard	Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
usb_disable_devices	Policy setup to disable per USB's PnP ID. There is no default value. Each device should be separated by a space. For example, **usb_disable_devices=0781:000c 0770:1234**.		Decoder
usb_enable_classes	Policy setup to enable per USB classes. There is no default value. Each class should be separated by a space. For example, **usb_enable_classes=audio video printer**.	same as "usb_disable_classes"	Decoder
usb_enable_devices	Policy setup to enable per USB's PnP ID. There is no default value. Each device should be separated by a space. For example, **usb_enable_devices=0781:000c 0770:1234**.		Decoder

KMoIP:

Key	Description	Value (bold is default)	Encoder/Decoder
kmoip_token_interval	It is a per decoder setting. Set the idle timeout value (in milliseconds) for holding an access token. For example, the default value is 100ms, which means once a decoder A acquired keyboard/mouse access, other decoders can only successfully acquire the access (token) after decoder A idle longer than 100ms. See "Application Guide - KMoIP" section for details. Normally, you don't need to set this value (leave it as default). It is used to workaround bad mouse, which keep reporting pointer activities and hence blocks all other decoders keyboard/mouse input.	**100**	Decoder
kmoip_roaming_layout	Enable and configure "KMoIP Multi-Screen Roaming" feature. See "KMoIP Multi-Screen Roaming" for details. Syntax: {*MAC*},{*X*},{Y}[:{*MAC*},{*X*},{Y}:...] Up to 16 "MAC,X,Y" pairs.	Default: "" (Empty string, means disable.)	Decoder
no_kmoip	Enable/disable KMoIP function.	y: disable KMoIP	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
		n: enable KMoIP	
osd_menu_font_size	The OSD menu font size	small **medium** large	
osd_menu_hot_key	OSD menu hot key, triple click for trigging the OSD menu for routing change on OSD menu	0: CAPS **1: CTL** 2: TAB 3: SHIFT	Decoder
osd_menu_position	OSD menu position on display.	**top_left** top_center top_right bottom_left bottom_center bottom_right left center right	Decoder
osd_menu_time_out_sec	OSD menu timeout, when there is no keyboard operating, timeout for canceling the OSD menu display.	0 ~ 3600 (**default: 10**) 0 means never	

Audio over IP:

Key	Description	Value (bold is default)	Encoder/Decoder
a_addon	Audio addon selection.	**none**	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
		dante (FW > 1.5.x) aes67(FW > 1.8.x)	
a_analog_in_vol	Set analog audio's input volume. Range from 0 to 100 %. Default value '-1' means use driver built-in default value. This setting requires system reset to take effect. To runtime change the volume, please use following command: **echo {*value*} > /sys/devices/platform/1500_i2s/analog_in_vol**	**-1**,0,1,2 ~,100	Encoder/Decoder
a_analog_out_vol	Set analog audio's output volume. Range from 0 to 100 %. Default value '-1' means use driver built-in default value. This setting requires system reset to take effect. To runtime change the volume, please use following command: **echo {*value*} > /sys/devices/platform/1500_i2s/analog_out_vol**	**-1**,0,1,2 ~,100	Encoder/Decoder
a_bridge_en	Enable bridge between two audio networks. (FW > 1.5.x)	**n** y	Encoder/Decoder
a_direction	Set analog audio as input or output, for encoder with programable analog audio connector	in **out**	Encoder
a_io_select	Encoder only.	**auto**: auto select analog when encoder's analog line in plugged (3.5mm socket) hdmi: Fixed hdmi audio analog: Fixed analog audio	Encoder

Key	Description	Value (bold is default)	Encoder/Decoder
a_out_src_sel	Source selection for I2S output. (FW > 1.5.x)	**native** addon	Decoder
addon_ip_mode	Audio Addon IP mode (Reboot required)	**dhcp** static	Encoder/Decoder
addon_ipaddr	Audio Addon IP Address (Reboot required)	IP Address	Encoder/Decoder
addon_netmask	Audio Addon IP netmask (Reboot required)	Netmask	Encoder/Decoder
addon_port	Assign Audio Addon to specific Ethernet port. Note: When addon_port used, dante_port will be ignored. Note: It will be only available for SKU has at least two separate Ethernet port (Reboot required)	**eth0** eth1	Encoder/Decoder
aes67_session_name	Session name on AES67 Transmitter	**'Same as hostname'**	Encoder/Decoder
aes67_ip	AES67 transmitting IP adder, only multicast IP address support now	**'Self-generated'**	Encoder/Decoder
aes67_port	AES67 transmitting port	**5004**	Encoder/Decoder
dante_ip_mode	Dante IP mode (Reboot required)	**dhcp** static	Encoder/Decoder
dante_ipaddr	Dante IP Address (Reboot required)	IP Address	Encoder/Decoder
dante_netmask	Dante IP netmask (Reboot required)	Netmask	Encoder/Decoder
dante_op_mode	Dante AV-A operation mode (Reboot required)	**AV-A** audio-only	Encoder/Decoder
dante_port	Assign Dante to specific Ethernet port. (Reboot required)	eth0 **eth1**	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
dante_trial_mode	Enable Dante trial mode without activation (one hour for free testing)	y **n**	Encoder/Decoder
i2s_clock_lock_mode	Audio genlock configuration. Default is enabled audio genlock. Disable it if compatibility issue occurs. This value is a 32bits hex value and upper 16bits is for gen-lock fine tune ppm range. Ex: 500ppm: 0x01F40000 100ppm: 0x00640000 50ppm: 0x00320000 25ppm: 0x00190000 Default value is 10ppm. Set ‘0’ will also apply default value.	**0: Enable genlock** 1: Normal 2: Low 4: High 0x80: Disable genlock	Decoder
no_i2s	Enable/Disable Audio over IP feature.	y: disable **n**: enable	Encoder/Decoder

ARP over IP (Audio Return Path over IP, IP5100 only)

Key	Description	Value (bold is default)	Encoder/Decoder
a_arp_input	Input source selection.	**toslink**: use Toslink interface hdmiout: use HDMI interface	Decoder
a_arp_no_earc	Force audio path arbitrator to ignore the selection of eARC.	**n: No effect** y: Ignore eARC	Encoder/Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
a_arp_publish_on	Option to turn on decoder ARP publish service.	**y**: enable ARP publisher n: disable ARP publisher	Decoder
a_arp_rr	Option to turn on decoder ARP subscribe service. New feature for decoder to subscribe another ARP published stream.	y: enable ARP subscriber **n**: disable ARP subscriber	Decoder
no_arp	Enable/disable ARP feature. Enable ARP must make sure i2s function (“no_i2s”) is also enabled.	y: disable ARP **n**: enable ARP	Encoder/Decoder

IR over IP:

Key	Description	Value (bold is default)	Encoder/Decoder
ir_guest_on	Enable IR guest mode feature	**y: Enable** n: Disable	Encoder/Decoder
ir_sw_decode	Enable IR software decode feature	y: Enable **n: Disable**	Encoder/Decoder
ir_sw_decode_nec_cfg	NEC remote control configuration of IR software decode feature The format is: {*device address*}_{*code for 0*}_{*code for 1*}_{*code for 2*}_.._{*code for 8*}_{*code for 9*} each variable is **decimal number** and valid device address is 0~65535 If device address is 65535 (default), IR software decode function only displays decoded NEC result includes device address and button code. This is for remote control learning purpose.	**65535_0_1_2_3_4_5_6_7_8_9**	Decoder

Key	Description	Value (bold is default)	Encoder/Decoder
no_ir	Enable/Disable IR over IP feature.	y: Disable **n**: Enable	Encoder/Decoder

Serial over IP:

Key	Description	Value (bold is default)	Encoder/Decoder
no_soip	Enable/Disable Serial over IP feature.	y: Disable **n**: Enable	Encoder/Decoder
s0_baudrate	Format: {*BAUDRATE*}-{*DATABITS*}{*PARITY*}{*STOPBITS*}	**115200-8n1**	Encoder/Decoder
soip_feedback_hex	Serial feedback data describe string as HEX mode or printable ASCII mode.	**y**: HEX n: Printable ASCII	Encoder/Decoder
soip_feedback_mode	Feedback mode is get serial data to multicast group to controller Pass-through is transmitting serial data between encoder and decoder	**y**: Enable for feedback n: Disable for pass-through	Encoder/Decoder
soip_feedback_wait	A timeout in ms for a serial data collection	**500**	Encoder/Decoder
soip_feedback_ip	Feedback report IP address	**226.2.0.0**	Encoder/Decoder
soip_feedback_port	Feedback report UDP port	**10000**	Encoder/Decoder
soip_guest_on	Working as gateway between TCP and serial.	y: Enable **n**: Disable	Encoder/Decoder
soip_type	Specify Serial over IP operating type.	1: Use type 1 (Legacy) **2**: Use type 2 3: Use type 3	Encoder/Decoder
soip_type2_token_timeout	Same for type 2 and type 3.	**1**,2,3... seconds	Encoder/Decoder

CEC over IP:

Key	Description	Value (bold is default)	Encoder/Decoder

cec_arc_proxy	The audio return channel (ARC) Proxy is a daemon to connecting (TV / Audio system) ARC automatically.
	On **encoder side**. ARC proxy is **analogy to a TV device** and if an audio system device (amplifier or sound bar) exists. Then the ARC proxy will try to open the ARC with audio system by CEC and reply to messages like a TV.
	On **decoder side**. ARC proxy is **analogy to an audio system device** and if TV device exists. Then the ARC proxy will try to open the ARC with TV by CEC and reply messages like an audio system.
	-
	About the rule of default value:
	The pre-condition is <In multicast mode> and <Board support ARC> and then
	if IS_HOST==y:
	enable_arc_proxy=y
	Else:
	if A_ARP_INPUT==hdmiout:
	enable_arc_proxy=y
	others
	enable_arc_proxy=n
	This feature usually uses with ARP.
		Default value:
		multicast_on = y: **y**
		multicast_on = n: **n**
			Encoder/Decoder

cec_arc_proxy_name	An OSD name (14 ASCII char within range [0x20..0x7E]) for TV/AudioSys to reply the CEC command <GIVE OSD NAME>.	Default value: Encoder: **TV** Decoder: **AV Receiver**	Encoder/Decoder

<table>
<tr><th>Key</th><th>Description</th><th>Value (bold is default)</th><th>Encoder/Decoder</th></tr>
<tr><td>cec_forward_rule</td><td>The arguments are:
1. 0xFF pass through
2. 0x00 block all
Below whitelist can bitwise by |
3. 0x01 allow "system audio control"
4. 0x02 allow "remote control pass through"
5. 0x04 allow "system standby"
6. 0x08 allow "one touch play"

The detail of each category:
System Audio Control:
Give Audio Status(71), Give System Audio Mode Status(7D), Report Audio Status(7A), Set System Audio Mode(72), System Audio Mode Request(70), System Audio Mode Status(7E).

Remote Control Pass Through:
User Control Pressed(44), User Control Released(45)

System Standby:
Standby(36)

One Touch Play:
Active Source(82), Image View On(04), Text View On(0D)</td><td>Default value:
multicast_on = y: 0x0
multicast_on = n: 0xff</td><td>Encoder/Decoder</td></tr>
<tr><td>no_cec</td><td>Enable/Disable CEC over IP feature.</td><td>y: Disable CECoIP
n: Enable CECoIP</td><td>Encoder/Decoder</td></tr>
</table>

Key	Description	Value (bold is default)	Encoder/Decoder
	CEC extension and guest mode are both activated if CECoIP is enabled.		

802.1X:

Key	Description	Value (bold is default)	Encoder/Decoder
ieee802_1x_enable	Enable 802.1X authentication.	y: enable **n**: disable	Encoder/Decoder
ieee802_1x_mode	802.1X authentication method.	peap|mschapv2 tls	Encoder/Decoder
ieee802_1x_mschapv2_password	Password for mschapv2 Note: You can directly use plain text password here or encrypt the password by using `wpa_encrypt -e <password>`		Encoder/Decoder
ieee802_1x_mschapv2_user	Username for mschapv2		Encoder/Decoder
ieee802_1x_tls_private_key_password	Password of the client private key Note: 1. Need to configure only if the client private key is encrypted by the password. 2. You can directly use plain text password here or encrypt the password by using `wpa_encrypt -e <password>`		Encoder/Decoder
ieee802_1x_tls_user	Username for TLS		Encoder/Decoder
ieee802_1x_validate_srv_ca	Validate authentication server certificate.	y: enable **n**: disable	Encoder/Decoder

Sink Power Control:

Key	Description	Value (bold is default)	Encoder/Decoder
cec_poweron_cmd	CEC power on command.	**40 04**	Decoder
cec_standby_cmd	CEC standby command.	**ff 36**	Decoder
ir_poweron_cmd	IR power on command.	N/A	Decoder
ir_standby_cmd	IR standby command.	N/A	Decoder
rs232_hex_cmd_enable	RS232 power command hex mode	y: Send RS232 command as hex encoded. **n**: Send RS232 command as ascii characters directly.	Decoder
rs232_param	RS232 power configuration. Format: {*BAUDRATE*}-{*DATABITS*}{*PARITY*}{*STOPBITS*}	**115200-8n1**	Decoder
rs232_poweron_cmd	RS232 power on command.	N/A	Decoder
rs232_standby_cmd	RS232 standby command.	N/A	Decoder
ir_poweron_cmd	IR power on command.	N/A	Decoder
ir_standby_cmd	IR standby command.	N/A	Decoder

Appendix B. List of Link Manager Events

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
e_802_1x	Take 802.1X setup effect	Encoder/ Decoder	**e_e_802_1x**	N/A	ieee802_1x_enable ieee802_1x_mode ieee802_1x_mschapv2_user ieee802_1x_mschapv2_password ieee802_1x_tls_user ieee802_1x_tls_private_key_pass word ieee802_1x_validate_srv_ca
e_a_direction	Set analog audio direction with in/out	Encoder	**e_e_a_direction::**{*DIRECTION*}	*DIRECTION:* > in > out	a_direction
e_a_input	Set audio input source selection	Encoder	**e_e_a_input**::{*INPUT*}	*INPUT:* > auto: Auto select input audio with plug GPIO detection.. > hdmi: Force select input with HDMI. > analog: Force select input with analog.	a_io_select
e_a_out_src_sel	Set audio output source selection on decoder	Decoder	**e_e_a_out_src_sel**::{*SOURCE*}	SOURCE: > native: native audio stream > addon: addon audio stream, could be Dante or AES67, pending on which addon is enabled, please refer astparam	a_out_src_sel

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
				'a_addon'	
e_addon_ethernet	Set Audio Addon to one of LAN ports	Encoder/ Decoder	**e e_addon_ethernet**::*{PORT}*	PORT: > eth0: LAN 1 > eth1: LAN 2	addon_port
e_adj_hdcp	Take HDCP setup effect	Encoder	**e e_adj_hdcp**	N/A	hdcp_enable
e_aes67_cli_start	Play the audio from a AES67 transmitter with SAP support.	Encoder/ Decoder	**e e_aes67_cli_start::***{IPADDR}*	*IPADDR*: IP address of source IP address of AES67 transmitter.	N/A
e_aes67_cli_stop	Stop playing the audio from a AES67 transmitter with SAP support.	Encoder/ Decoder	**e e_aes67_cli_stop**		N/A
e_aes67_session	Setup AES67 Transmitter session info	Encoder/ Decoder	**e e_aes67_session**::*{NAME}*::*{IP}*::*{PORT}*	NAME: AES67 SDP Session Name IP: AES67 Transmitting IP address, only support multicast IP address PORT: AES67 Transmitting Port	aes67_session_name aes67_ip aes67_port
e_analog_out_volume	Set analog audio output volume	Encoder/ Decoder	**e e_analog_out_volume**::{*VOLUMN*}	*VOLUMN:* The analog audio's output volume, from **0** to **100** %	a_analog_out_vol
e_arp_hdmiout_cap	Set HDMIOUT ARP capability (eARC not available yet)	Decoder	**e e_arp_hdmiout_cap**::{*CAPABILITY*}	*CAPABILITY*: > arc: HDMI ARC only > earc: both HDMI eARC & ARC	a_arp_hdmiout_cap
e_arp_input	Set ARP input	Decoder	**e e_arp_input**::{*INPUT*}	*INPUT*: > toslink > hdmiout	a_arp_input
e_cec_one_touch_play	Send CEC One Touch Play command	Decoder	**e e_cec_one_touch_play**	N/A	N/A

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
e_cec_system_standby	Send CEC System Standby command	Decoder	**e_e_cec_system_standby**	N/A	N/A
e_discovery	Set discovery hostname	Encoder/ Decoder	**e_ e_discovery**::{*HOSTNAME*}::{*BONJOUR_EN*}::{*SSDP_EN*}::{*LLDP_EN*}	*HOSTNAME*: Hostname for discovery *BONJOUR_EN*: > y: enable Bonjour service. > n: disable Bonjour service. *SSDP_EN*: > y: enable SSDP service. > n: disable SSDP service. *LLDP_EN*: > y: enable LLDP service. > n: disable LLDP service.	changed_os_hostname bonjour_enable ssdp_enable lldp_enable
e_edid_fast_switch_mode	Set EDID patch for fast switching, will reduce current and any future EDID Extended Color, HDR, 4:2:0 support.	Encoder	**e_e_edid_fast_switch_mode**::{*MODE*}	*MODE*: > y: enable the EDID fast switch mode. > n: disable the EDID fast switch mode	edid_fast_switch_mode
e_enable_usb_b_network	Enable/Disable USB-B Ethernet feature	Encoder	**e e_enable_usb_b_network**::*{ENABLE}*	*ENABLE*: > y: enable USB-B Network. > n: disable USB-B Network.	en_usb_b_network
e_enable_usb_c_network	Enable/Disable USB-C Ethernet feature	Encoder	**e e_enable_usb_c_network**::*{ENABLE}*	*ENABLE*: > y: enable USB-C Network. > n: disable USB-C Network.	en_usb_c_network
e_find_me	Set Status LED flashing very fast for find the unit	Encoder/ Decoder	**e_e_find_me**::{*TIME*}	*TIME*: How long time in seconds for the STATUS LED flashing with 125 ms on / 125ms off frequency.	N/A

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
				Range: 0 ~, 0 means stop flashing immediately.	
e_video_genlock_scaling	Set genlock scaling mode	Decoder	**e_ e_video_genlock_scaling**::{*ENABLE*}	*ENABLE:* > y: enable genlock scaling > n: disable genlock scaling	video_genlock_scaling
e_hdmi_audio_mute	Mute HDMI Audio	Decoder	**e e_hdmi_audio_mute**::{*MUTE*}	MUTE: > y: mute > n: unmute	hdmi_audio_mute
e_lan_mode	Set Dual RJ45 LAN ports product LAN working mode	Encoder/ Decoder	**e e_lan_mode**::*{MODE}*	MODE: > switched (default): LAN ports working as regular network switch, LAN 1, LAN 2, LAN 3 and CPU are connected > bridged: LAN 2 port is bridged to LAN 1 and LAN 3 ports, not connected to CPU > independent: LAN 2 port is independent with LAN 1 and LAN 3, LAN 2 can be assigned with Dante / AES67 only.	lan_mode
e_ntp_off	Disable NTP client	Encoder/ Decoder	**e e_ntp_off**		enable_ntp_cli
e_ntp_on	Enable NTP client	Encoder/ Decoder	**e e_ntp_on**::{*NPTADDR*}	*NTPADDR*: The IP address or URL of NTP server	enable_ntp_cli ntp_srv_addr
e_osd_off_str	Disable showing string on GUI	Decoder	**e e_osd_off_str**::{*UUID*}	*UUID*: support "now" only	N/A
e_osd_on_str	Enable showing string on GUI	Decoder	**e_ e_osd_on_str**::{Y_START}::{Y_SIZE}::{TRANSPARENT}::{BG_MASK_EN}::{	*Y_START*: Y start point *Y_SIZE*: Pixels *TRANSPARENT*: Valid in range 0 ~ 31 (0:transparent ~ 31:opaque).	N/A

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
			OFF_TIMER}::{STR}::{FONT_SIZE}::{FONT_COLOR}	*BG_MASK_EN*: Enable or Disable color keying effect. > 0: opaque/disable color keying > 1: transparent/enable color keying *OFF_TIMER*: off timer in seconds, 'n' = always on *STR*: OSD string content *FONT_SIZE*: font size in pt *FONT_COLOR*: **MM**RRGGBB > MM: Alignment mode, should be set to 0xFF(Center align) or 0xF0 (Top-Left align). > RR: Red > GG: Green > BB: Blue	
e_osd_position	Set the OSD position on GUI	Decoder	**e_ e_osd_position**::{*ALIGN*}::{*X_START_OFFSET*}::{*Y_START_OFFSET*}	*ALIGN*: > 0: Disable alignment overwrite. Use default behavior (Center). > 1: Top-Left > 2: Top > 3: Top-Right > 4: Left > 5: Center > 6: Right > 7: Bottom-Left	N/A

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
				> 8: Bottom > 9: Bottom-Right *X_START_OFFSET*: Extra horizontal offset from alignment. In pixels *Y_START_OFFSET*: Extra vertical offset from alignment. In pixels.	
e_osd_menu_font_size	Set OSD menu font size	Decoder	**e_e_osd_menu_font_size::** {*KEY_VAL*}	*SIZE:* > small > medium > large	osd_menu_font_size
e_osd_menu_hot_key	Set OSD menu hot key	Decoder	**e_e_osd_menu_hot_key**::{*KEY_VAL*}	*KEY_VAL:* > **-1:DISABLE** > 0:CAPS LOCK > 1:CTL > 2:TAB > 3:SHIFT	osd_menu_hot_key
e_osd_menu_position	Set OSD menu position	Decoder	**e_e_osd_menu_position::** {*POSITION*}	*POSITION:* > top_left > top_center > top_right > bottom_left > bottom_center > bottom_right > left > center	osd_menu_position

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
				> right	
e_osd_set_menu_time_out_sec	Set OSD menu timeout without activity	Decoder	**e_e_osd_menu_time_out_sec::**{*TIMEOUT*}	*TIMEOUT:* 0~3600 seconds	osd_menu_time_out_sec
e_reconnect	Reconnect the feature to encoder / decoder	Encoder/ Decoder	**e_ e_reconnect**{::*chselect*}{**::**{**z**|**Z**}|{**v**||**u**||**a** ||**r**||**s**||i}}	*chselect*: > the MAC address without 'colon' as channel ID > number from 0000 to 9999 > For decoder it could be the IP address of encoder {z|Z}|{v||u||a||r||s||i}: > v: Video over IP > u: USB over IP > a: Audio over IP > r: IR over IP > s: Serial over IP > i: Audio Return over IP, for encoder > z: all features. == **vuasr** > Z: all features except video over IP. == **uasr** NOTE: > **can be any combination and order** which indicate connection order of specified service. > **'i'** for encoder only.	N/A

<table>
<tr><th>Function</th><th>Description</th><th>Models</th><th>Syntax</th><th>Parameters/Attributes</th><th>Related astparam</th></tr>
<tr><td>e_set_account</td><td>Set the Console API account and password</td><td>Encoder/ Decoder</td><td>e_
e_set_account::{USER}::{PASSWORD}::{ROLE}</td><td>USER: username
PASSWORD: Passwrod encrypted by DES.
ROLE:
> admin: root level
> user: not support yet (not support yet)</td><td>account_info</td></tr>
<tr><td>e_set_aesmode</td><td>Set the AES mode for stream encryption</td><td></td><td>e_e_set_aesmode::{MODE}</td><td>MODE:
> 0: Standard mode (No Jumbo Required)
> 1: Super mode(Jumbo Required)</td><td>v_eng_drv_option</td></tr>
<tr><td>e_set_aeskey</td><td>Set the AES key for stream encryption</td><td>Encoder/ Decoder</td><td>e_e_set_aeskey::{KEY}</td><td>KEY:
> 128 bit key described by 16 bytes hex string
> 0: chip build-in default key
> 1: disable AES128 encryption</td><td>aes_key</td></tr>
<tr><td>e_set_time_zone</td><td>Set time zone</td><td>Encoder/ Decoder</td><td>e_e_set_time_zone::{TIMEZONE}</td><td>TIMEZONE:
Time zone name, could be read from http(s)://ip_addr/device/zoneinfo</td><td>time_zone</td></tr>
<tr><td>e_stop_link</td><td>Stop the feature on encoder / decoder</td><td>Encoder/ Decoder</td><td>e_e_stop_link{::{z|Z}|{v||u||a||r||s||i}}</td><td>Reference 'e_reconnect'</td><td>N/A</td></tr>
<tr><td>e_switch_board_type</td><td>Runtime switching IPT5100U board type</td><td>IPT5100 U</td><td>e_e_switch_board_type::{TYPE}</td><td>TYPE:
> host: TX / encoder mode
> client: RX / decoder mode</td><td>usb_board_type</td></tr>
<tr><td>e_usb_b_ethernet</td><td>Set USB-B Ethernet LAN port</td><td>Encoder/ Decoder</td><td>e e_usb_b_ethernet::{PORT}</td><td>TYPE: When lan_mode is set as independent, which port USB-B Ethernet will be assigned
> lan1: LAN 1
> lan2: LAN 2</td><td>usb_b_ethernet</td></tr>
</table>

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
e_usb_c_ethernet	Set USB-C Ethernet LAN port	Encoder/ Decoder	**e e_usb_c_ethernet**::*{PORT}*	*TYPE*: When lan_mode is set as independent, which port USB-C Ethernet will be assigned > lan1: LAN 1 > lan2: LAN 2	usb_c_ethernet
e_usbc_charger	Turn on/off the USB-C charger	IPE5100 W-000/B70	**e e_usbc_charger::**{*ENABLE*}	*ENABLE*: > y: Turn on the USB-C Charger > n: Turn off the USB-C Charger, it's the default too.	usbc_charger Example syntax: **astparam g usbc_charger**
e_v_switch_in	Manually switch input to specific port	Switch Encoder	**e e_v_switch_in::**{*PORT*}	*PORT*: > hdmi1 > hdmi2 > usb-c	v_switch_in
e_v_switch_mode	Set the switch mode	Switch Encoder	**e e_v_switch_mode::**{*MODE*}	*MODE*: > manual > priority > auto	v_switch_mode
e_v_switch_pri	Set the switch priority when switch mode is priority	Switch Encoder	**e e_v_switch_pri::** {*PORT*}**::**{*PORT*}**::**{*PORT*}	*PORT*: > hdmi1 > hdmi2 > usb-c	v_switch_pri
e_vw_enable	Set the video wall configuration	Decoder	**e e_vw_enable**_{M}_{*N*}_{*X*}_{*Y*}[_{*V*}]	Please reference 'Video Wall Settings'	vw_ver vw_v2_x1 vw_v2_y1 vw_v2_x2 vw_v2_y2

Function	Description	Models	Syntax	Parameters/Attributes	Related astparam
					vw_max_row vw_max_column vw_row vw_column
e_vw_force_refresh	Force refresh the video wall configuration from flash parameter update	Decoder	**e_e_vw_force_refresh**	N/A	N/A
e_vw_moninfo	Set the video wall monitor info for bezel compensation of video wall APIv1	Decoder	**e_** **e_vw_moninfo**_*{HA}_{HT}_{VA}_{VT}*	*HA*: Horizontal active area of monitor in mm *HT*: Horizontal total area of monitor in mm *VA*: Vertical active area of monitor in mm *VT*: Vertical total area of monitor in mm	vw_moninfo_ha vw_moninfo_ht vw_moninfo_va vw_moninfo_vt
e_vw_rotate	Set the monitor rotation	Decoder	**e_e_vw_rotate**_{*ROTATE*}	*ROTATE*: > 0: No rotate > 1: vertical flip > 2: horizontal flip > 3: 180 degree > 5: 90 degree (IPD5100) > 6: 270 degree	vw_rotate
e_vw_stretch_type	Set the stretch mode from show original video to connected display	Decoder	**e_e_vw_stretch_type**_{*STRETCH* }	*STRETCH*: > 1: Stretch out > 2: Fit in	vw_stretch_type
e_web_enable	Enable or disable web service	Encoder/ Decoder	**e_e_web_enable**::{*ENABLE*}	*ENABLE*: > on: Enable web service. > off: Disable web service	web_enable

Appendix C. List of lmparam

Most of lmparam are not allowed to be written from user. Unless special noted, write to lmparam may cause unexpected result. User should use it only for querying link manager's internal flags.

Common

lmparam	Description
STATE	Link Manager State. Encoder: > s_init: System is under initialization. > s_idle: System is idle. Services are stopped. > s_attaching: Services are started, but no decoder attached or no video source. > s_srv_on: Services are started. > s_error: Fatal error. Decoder: > s_init: System is under initialization. > s_idle: System is idle. Services are stopped. > s_srv_on: Services are started. > s_error: fatal error.
IP_MODE	Maps from astparam, ip_mode.

MY_IP	IP address.
MY_MAC	Ethernet MAC address.
ETH_LINK_STATE	Ethernet PHY link status.
ETH_LINK_MODE	Ethernet PHY link mode. Only valid when 'ETH_LINK_STATE' is on.
FREE_ROUTING	Maps from astparam, free_routing.
MCIP_SRV_MAP	Maps from astparam, mcip_srv_map.

Encoder

Imparam	Description
CH_SELECT_I	Channel selection for ARP (Audio Return Path)
V_QUALITY_MODE	Current video quality mode
V_BCD_THRESHOLD	Current video Anti-Dither mode.
USB_CLIENT_IP	Current USB decoder IP address.

Decoder

Imparam	Description
CH_SELECT CH_SELECT_V CH_SELECT_U CH_SELECT_A	Channel selection. In decoder, each service can choose it own channel to connect to. It is so called 'free routing'. CH_SELECT will be used if not specified.

CH_SELECT_R CH_SELECT_S	
U_USBIP_STATE	Current USBoIP status. Get by ulmparam.
U_KMOIP_STATE	Current KMoIP status. Get by ulmparam.
GWIP	Encoder’s IP address that decoder’s video over IP is connected/connecting to.

Appendix D. List of Driver sysfs

Function	Description	Related lmparam	Related astparam	Encoder/Decoder
Video Quality Mode	echo {*QUALITY_MODE*} > /sys/devices/platform/videoip/QualityMode	V_QUALITY_MODE	ast_video_quality_mode	Encoder/Decoder
Frame Rate Control	echo {*FRAME_RATE*} > /sys/devices/platform/videoip/frame_rate_control	V_FRAME_RATE	v_frame_rate	Encoder
Detach Input	echo {0|1} > /sys/devices/platform/videoip/detach_input_port	N/A	N/A	Encoder
EDID USE	echo {primary|secondary} > /sys/devices/platform/videoip/edid_use	EDID_USE	edid_use	Encoder/Decoder
Read EDID	cat /sys/devices/platform/videoip/edid_cache	N/A	N/A	Encoder
Write EDID	echo "{*EDID*}" > /sys/devices/platform/videoip/eeprom_content *EDID*: A hex described string with space or comma separated, and need . e.g. echo "00 ff ff ff ff ff ff 00 63 18 00 00 82 00 00 00 20 19 01 03 80 59 32 78 0a ee 91 a3 54 4c 99 26 0f 50 54 21 08 00 81 c0 81 00 81 80 90 40 95 00 b3 00 01 00 01 01 04 74 00 30 f2 70 5a 80 b0 58 8a 00 c4 8e 21 00 00 1e 02 3a 80 18 71 38 2d 40 58 2c 45 00 50 1d 74 00 00 1e 00 00 00 fd 00 18 3c 1e 5a 1e 00 0a 20 20 20 20 20 20 00 00 00 fc 00 48 44 4d 49 0a 20 20 20 20 20 20 20 20 01 71 02 03 38 f1 4e df 10 1f 04 13 20 21 22 5d 5e 5f 01 12 03 23 09 07 07 83 01 00 00 6d 03 0c 00 10 00 b8 3c 21 00 60 01 02 03 e2 00 0f e3 0e 60 61 e3 05 03 01 e3 06 05 01 01 1d 00 72 51 d0 1e 20 6e 28 55 00 60 59 21 00 00 1e 66 21 56 aa 51 00 1e 30 46 8f 33 00 50 1d 74 00 00 1e 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 8a" > /sys/devices/platform/videoip/eeprom_content	N/A	N/A	Encoder
Timing	cat /sys/devices/platform/videoip/timing_info	N/A	N/A	Encoder/Decoder
State	cat /sys/devices/platform/videoip/State	N/A	N/A	Encoder/Decoder
Monitor Info	cat /sys/devices/platform/display/monitor_info	N/A	N/A	Decoder

Function	Description	Related lmparam	Related astparam	Encoder/Decoder
Monitor EDID	cat /sys/devices/platform/display/monitor_edid	N/A	N/A	Decoder
Shutdown HDMI Out	echo {1|0} > /sys/devices/platform/display/screen_off	N/A	N/A	Decoder
Output Timing	echo {*TIMING*} > /sys/devices/platform/videoip/output_timing_convert	V_OUTPUT_TIMING_CONVERT	v_output_timing_convert	Decoder
Input Audio Info	cat /sys/devices/platform/1500_i2s/input_audio_info	N/A	N/A	Encoder
Output Audio Info	cat /sys/devices/platform/1500_i2s/output_audio_info	N/A	N/A	Decoder

Appendix E. List of other Commands

Function	Description	Syntax	Parameters/Attributes	Encoder/Decoder
cec_send	CEC frames generation.	cec_send {*FRAMES*}	*FRAMES*: A string with hexadecimal numbers joined with colon or space. e.g. cec_send 4f:36 cec_send 40 04	Encoder/Decoder
gbconfig	A legacy command, please refer Syntax and Parameters/Attributes	gbconfig {[*OPTS*]=[*VAL*]}	[*OPTS*]=[*VAL*]: --hdcp-enable={y|n} Set encoder HDMI Input's HDCP capability, when 'y' set, the encoder will support up to HDCP2.3 --source-select={*CHSELECT*} Same as **e_e_reconnect::**{*chselect*} --source-select-v={*CHSELECT*} Same as **e_e_reconnect::**{*chselect*}::v --source-select-a={*CHSELECT*} Same as **e_e_reconnect::**{*chselect*}::a --source-select-u={*CHSELECT*} Same as **e_e_reconnect::**{*chselect*}::u --source-select-r={*CHSELECT*} Same as **e_e_reconnect::**{*chselect*}::r --source-select-s={*CHSELECT*} Same as **e_e_reconnect::**{*chselect*}::s --sinkpower-ir={y|n} Enable or disable IR signal for 'sinkpower' command	Encoder/Decoder

Function	Description	Syntax	Parameters/Attributes	Encoder/Decoder
			--sinkpower-rs232={y|n} Enable or disable RS232 signal for 'sinkpower' command --sinkpower-cec={y|n} Enable or disable CEC signal for 'sinkpower' command --rs232-param={*RS232PARAM*} Set RS232 parameters for 'sinkpower' command --rs232-hex-cmd-enable={y|n} Set RS232 describe with HEX string for 'sinkpower' command --soip-feedback-mode={y|n} Enable or disable soip feedback mode --line-in --level-control={*VOLUME*} Set line in volume from 0 ~ 100, 0 will mute. --line-in --level-up Set line in volume up with 5%. --line-in --level-down Set line in volume down with 5%. --line-out --level-control={*VOLUME*} Set line out volume from 0 ~ 100, 0 will mute. --line-out --level-up Set line out volume up with 5%. --line-out --level-down Set line out volume down with 5%. --show Together with other parameters for obtain the value configured.	
gbstatus	get video / audio status	gbstatus	Response as below	Encoder/Decoder

Function	Description	Syntax	Parameters/Attributes	Encoder/Decoder
			Encoder: hdmi_in_active: {true|false} resolution: {*RESOLUTION*} hdmi_in_frame_rate: {*FPS*} audio_input_format: {none|hbr|nlpcm|lpcm} audio_bitrate: {*BITRATE*} hdcp: {none|1.x|2.2} color_space: {YCbCr|RGB} chroma_sampling: {4:4:4|4:2:2|4:2:0} color_depth: {8|10|12} hdr: {SDR|HDR10|Dolby_Vision_Low_Latency|Dolby_Vision_Starndard} *audio_sample_rate:* {22.05_KHz|24_KHz|32_KHz|44.1_KHz|48_KHz|88.2_KHz|96_KHz|176.4_KHz|192_KHz|768_KHz} Decoder: hdmi_out_active: {true|false} resolution: {*RESOLUTION*} hdmi_out_frame_rate: {*FPS*} audio_output_format: {none|hbr|nlpcm|lpcm} audio_bitrate: {*BITRATE*} hdcp: {none|1.x|2.2} color_space: {YCbCr|RGB} chroma_sampling: {4:4:4|4:2:2|4:2:0} color_depth: {8|10|12}	

Function	Description	Syntax	Parameters/Attributes	Encoder/Decoder
			hdr: {SDR|HDR10|Dolby Vision Low Latency|Dolby Vision Starndard} *audio sample rate:* {22.05 KHz|24 KHz|32 KHz|44.1 KHz|48 KHz|88.2 KHz|96 KHz|176.4 KHz|192 KHz|768 KHz}	
irs	IR generation.	irs {*IRCODE*}	*IRCODE*: Could be PRONTO HEX string or Global Cache digit format e.g. PRONTO: / # **irs 0000 0067 0023 0000 0158 00b4 0014 0017 0014 0017 0013 0017 0014 0017 0014 0017 0014 0017 0014 0017 0013 0017 0014 0043 0014 0044 0014 0043 0014 0043 0014 0044 0014 0043 0014 0043 0014 0044 0014 0017 0014 0043 0013 0017 0014 0017 0014 0043 0015 0016 0014 0017 0014 0017 0014 0043 0014 0017 0014 0043 0014 0044 0014 0017 0014 0043 0014 0043 0014 0044 0013 063a 0158 005a 0014** Global Cache: / # **irs 40000,1,1,344,180,20,23,20,23,20,23,20,23,20,23,20,23,20,23,20,23,20,67,20,68,20,67,20,67,19,68,20,67,20,67,20,68,20,23,20,23,20,67,20,67,20,67,20,23,19,23,20,23,20,67,20,68,20,23,19,23,20,23,20,67,20,67,20,68,20,1594,343,90,21,2006**	Encoder/Decoder
osd_on.sh	Show OSD string on display with default position	osd_on.sh "{*OSDSTRING*}"	*OSDSTRING*: A string with printable characters, support '\n' for making new line.	Decoder
osd_off.sh	Disable the OSD showing	osd_off.sh	Disable the OSD showing immediately.	Decoder
reboot	Reboot the device	reboot	N/A	Encoder/Decoder

Function	Description	Syntax	Parameters/Attributes	Encoder/Decoder
reset_to_default.sh	Reset to factory default, require 'reboot' for taking effect.	reset_to_default.sh	N/A	Encoder/Decoder
sinkpower	Operate sink power control	sinkpower {on|off}	> on: Operate RS232, CEC, IR command for turn on display, related on gbconfig configuration. > off: Operate RS232, CEC, IR command for turn off display, related on gbconfig configuration	Encoder/Decoder
soip2	Send data to serial when feedback mode is enabled.	soip2 -f /dev/ttyS0 -b {*RS232-PARAM*} [-H] [-r] [-n] -s "*CONTENT*"	*RS232-PARAM*: Format "b-dps" > b - baud rate > d - data bits > p - parity > s - stop bit -H: Identify the "CONTENT" is a hex format description string and update the feedback hex setting as same format to controller. -r: append <CR> to the endo of CONTENT -n: append <LF> after <CR> or to the end of CONTENT *CONTENT*: The serial content need be sent.	Encoder/Decoder

Appendix F. List of HTTP Commands

<table>
<tr><th>HTTP URL</th><th>Description</th><th>HTTP Method</th><th>HTTP Content-Type</th><th>Parameters/Attributes or Result</th><th>Encoder/Decoder</th></tr>
<tr><td>/upload_security</td><td>Upload 802.1X Cert Files</td><td>POST</td><td>multipart/form-data;</td><td><table><tr><th>Form Name</th><th>Form Value Type</th><th>Form Value</th></tr><tr><td>802_client</td><td>octet_stream</td><td>(binary)</td></tr><tr><td>802_client_key</td><td>octet_stream</td><td>(binary)</td></tr><tr><td>802_ca</td><td>octet_stream</td><td>(binary)</td></tr></table></td><td>Encoder/Decoder</td></tr>
<tr><td>/stream</td><td>The mjpeg preview stream</td><td>GET</td><td>N/A</td><td>MJPEG stream</td><td>Encoder/Decoder</td></tr>
<tr><td>/capture.jpg</td><td>The jpeg preview snapshot</td><td>GET</td><td>N/A</td><td>JPEG file</td><td>Encoder/Decoder</td></tr>
<tr><td>/cgi-bin/astfwup.cgi</td><td>Update the unit with firmware file</td><td>POST</td><td>multipart/form-data;</td><td><table><tr><th>Form Name</th><th>Form Value Type</th><th>Form Value</th></tr><tr><td>fwfile</td><td>octet_stream</td><td>(binary)</td></tr></table></td><td>Encoder/Decoder</td></tr>
<tr><td>/update_fw_info.txt</td><td>FW updates progress</td><td>GET</td><td>N/A</td><td>The FW update info, need polling it, content like below:
firmware file size: 8230925 bytes
Decompressing firmware...
Platform matched.
Start programming flash...
programming kernel...
programming rootfs...
Programming completed</td><td>Encoder/Decoder</td></tr>
<tr><td>/update_fw_result.txt</td><td>FW updates result</td><td>GET</td><td>N/A</td><td>The FW update result, need polling it, content like below:
When FW update start:
Please wait...</td><td>Encoder/Decoder</td></tr>
</table>

HTTP URL	Description	HTTP Method	HTTP Content-Type	Parameters/Attributes or Result	Encoder/Decoder
				When FW update done: DONE. Rebooting...	
/fw_size_total.js	FW updates total size	GET	N/A	The FW update total size, need polling it, content like below: FWSizeTotal = 9689419;	Encoder/Decoder
/fw_size_remain.js	FW updates remain size, calculate with total size will get the progress of FW updates.	GET	N/A	The FW update remain size, need polling it, content like below: FWSizeRemain = 6952751;	Encoder/Decoder
/cgi-bin/query.cgi	A CGI for apply Console API on HTTP way	GET	N/A	With HTTP query, format as: callback=*CALLBACKSTRING*&cmd=*CONSOLEAPI*&wrap_type=multi_line callback=*CALLBACKSTRING* It will reponse all output to a callback string body like 'callback()' cmd=*CONSOLEAPI* space should be replaced by '+' The command need be applied to board, response will be put to HTTP response wrap_type=multi_line without wrap_type, will output regular text with multi_line will convert linefeed with unicode translation	Encoder/Decoder
/device/log	Pack and download logs	GET	N/A	With HTTP query, format as: name=*FILENAME* If the name is specified, device will use the name as download name instead of self assigned name, because device may not have correct date and time configuration.	Encoder/Decoder

HTTP URL	Description	HTTP Method	HTTP Content-Type	Parameters/Attributes or Result	Encoder/Decoder

/device/zoneinfo	Download available zone info list	GET	N/A	A json file will be downloaded for get available zone list,
				'name' is used for showing
				'dst' is use for setting with api 'e_set_time_zone'
				{
				  "zoneinfo": [
				    {
				      "name": "-12:00 Etc/GMT+12",
				      "dst": "Etc/GMT+12"
				    },
				    ......
				  ]
				}
					Encoder/Decoder

Appendix G. Video Wall Settings

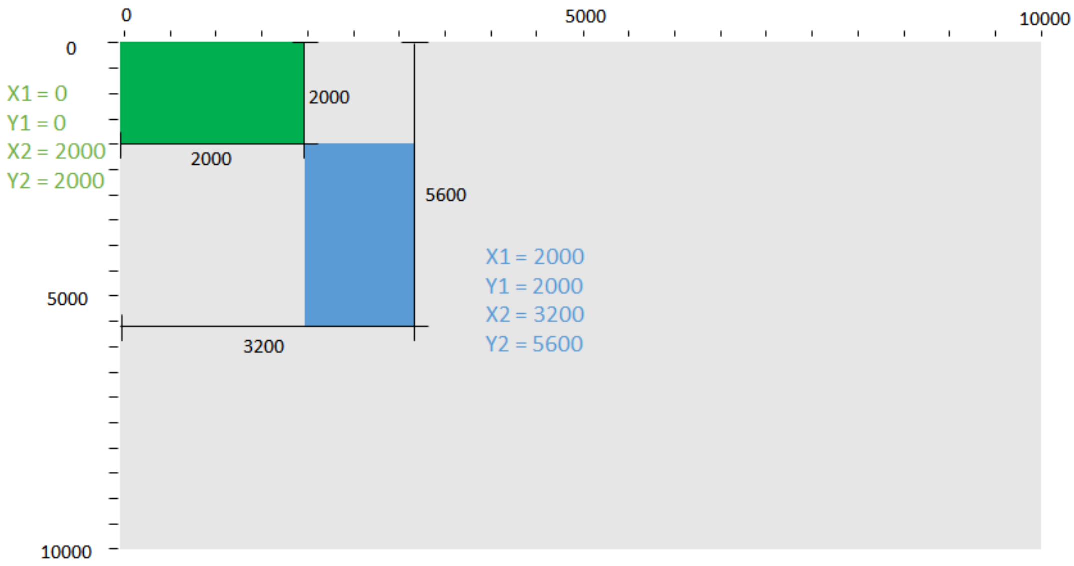
Concepts

We use 2 virtual coordinates, top left (x1, y1) and bottom right (x2, y2), to select the area of source image for scaling-up at decoder side.

The unit of x (x1 or x2) is in ‱ of width, range: 0 ~ 10000

The unit of y (y1 or y2) is in ‱ of height, range 0 ~ 10000

Example



Video Wall Console API Setup

V1: e_vw_enable_M_N_x_y or e_vw_enable_M_N_x_y_1

V2: e_vw_enable_X1_Y1_X2_Y2_2

V2: e_vw_rotate_ROTATE

For V2 API, the range for X1, Y1, X2, Y2 is: 0 ~ 10000, ROTATE range is 0, 3(clockwise 180 degree), 5(clockwise 90 degree), or 6(clockwise 270 degree)

There are several parameters to obtain the settings of video wall for APIv2

Key	Description	Value (Bold is default)
vw_rotate	Clockwise rotate output picture	**0**: No rotate 3: 180 degrees 6: 270 degrees

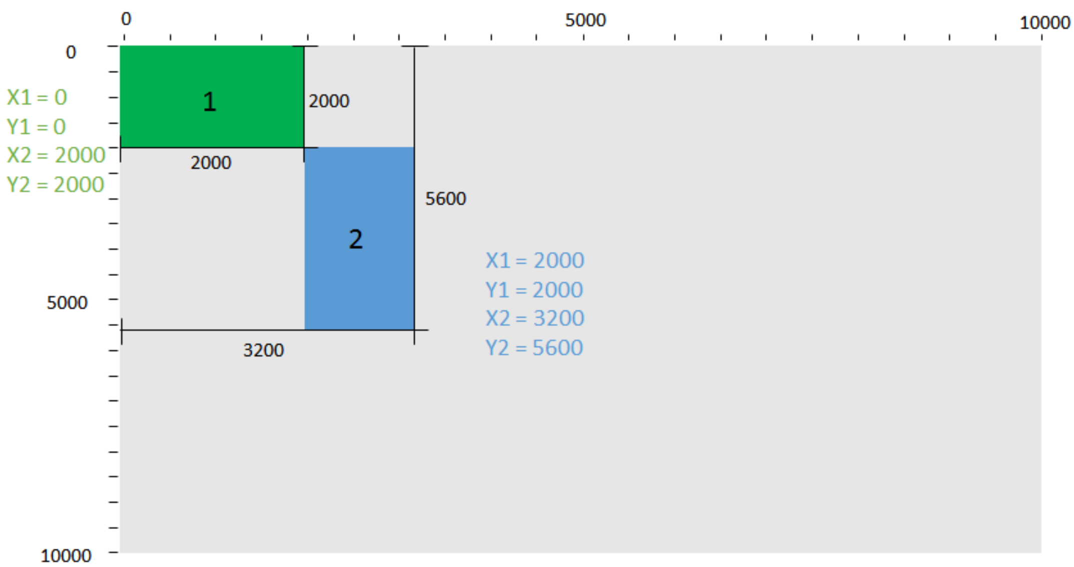
vw_ver	Specify video wall control APIs version: 1: Default. Specify monitor layout and auto calculate coordinate. 2: Manually specify coordinate. Easier for mosaic style video wall configuration.	**1**: Default 2: v2. Specify coordinate.
vw_v2_x1	x value of top left coordinate	**0**~10000
vw_v2_y1	y value of top left coordinate	**0**~10000
vw_v2_x2	x value of bottom right coordinate	**0**~10000
vw_v2_y2	y value of bottom right coordinate	**0**~10000

Example:

- Using API v2 to select an area span by virtual coordinates (5000, 5000, 10000, 10000) for scaling-up
  # e e_vw_enable_5000_5000_10000_10000_2
- Using API v2 to disable scaling up
  # e e_vw_enable_0_0_0_0_2
- Using API v1 to set decoder r0c1's video wall screen layout as 2x3's row 0 column 2.
  # e e_vw_enable_1_2_0_2 or
  # e e_vw_enable_1_2_0_2_1
- Set rotate type to 270 degrees.
  # e e_vw_rotate_6

Installation Procedure Using Console APIs

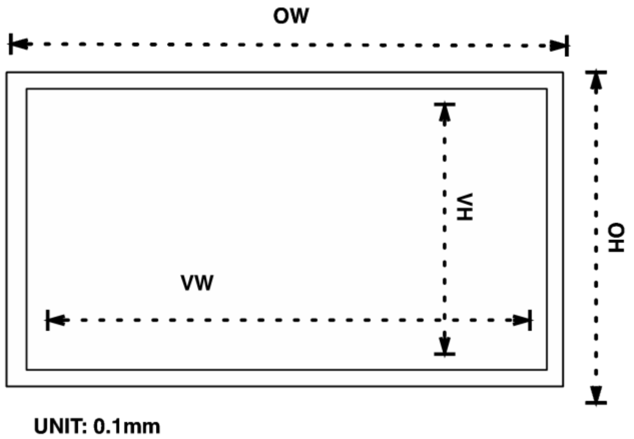
General Case



1 X1 = 0 Y1 = 0 X2 = 2000 Y2 = 2000 #e e_vw_enable_0_0_2000_2000_2	**2** X1 = 2000 Y1 = 2000 X2 = 3200 Y2 = 5600 and 270 degree clockwise rotate #e e_vw_enable_2000_2000_3200_5600_2 #e e_vw_rotate_6

MxN Video Wall

Consider about the border of display devices, the coordinates calculation should include the relations described below,



Following command example applies to MxN video wall with monitor dimension in video wall API v2,

e e_vw_enable_X1_Y1_X2_Y2_2

The decoders' position be:

R0C0	R0C1	...	R0Cn
R1C0	R1C1	...	R1Cn
...	...	...	...
RmC0	RmC1	...	RmCn

For the device located at row a, column b, RaCb:

X1' = b * OW

Y1' = a * OH

X2' = b *OW + VW

Y2' = a * OH + VH

Xtotal: n * OW + VW

Ytotal: m * OH + VH

X1 = SCALE * X1'/ Xtotal

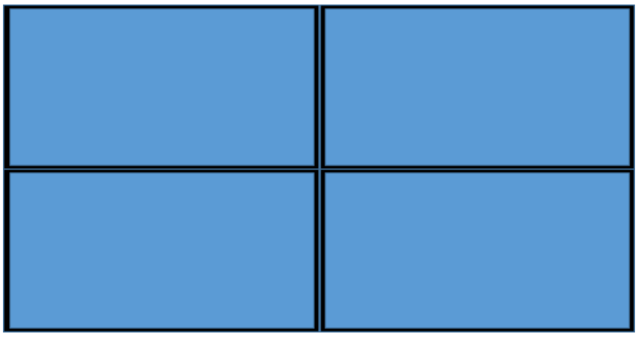
Y1 = SCALE * Y1' / Ytotal

X2 = SCALE * X2' / Xtotal

Y2 = SCALE * Y2' / Ytotal

a: 0 ~m, m=M-1; b: 0~n, n=N-1; SCALE=10000

2x2 Video Wall

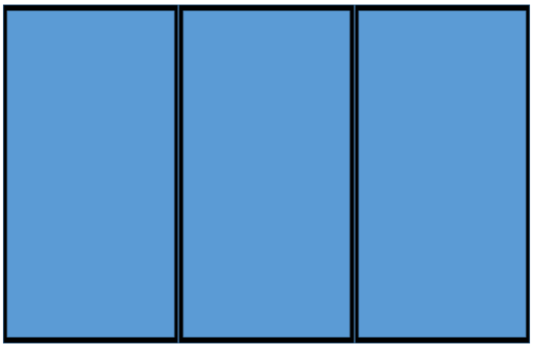


OW=200, VW=190, OH=100, VH=90

The decoders' position and commands will be:

r0c0: X1 = 0 Y1 = 0 X2 = 4871 Y2 = 4736 #e e_vw_enable_0_0_4871_4736_1	**r0c1:** X1 = 5128 Y1 = 0 X2 = 10000 Y2 = 4736 #e e_vw_enable_5128_0_10000_4736_1
r1c0: X1 = 0 Y1 = 5263 X2 = 4871 Y2 = 10000 #e e_vw_enable_0_5263_4871_10000_1	**r1c1:** X1 = 5128 Y1 = 5263 X2 = 10000 Y2 = 10000 #e e_vw_enable_5128_5263_10000_10000_1

1x3 Video Wall



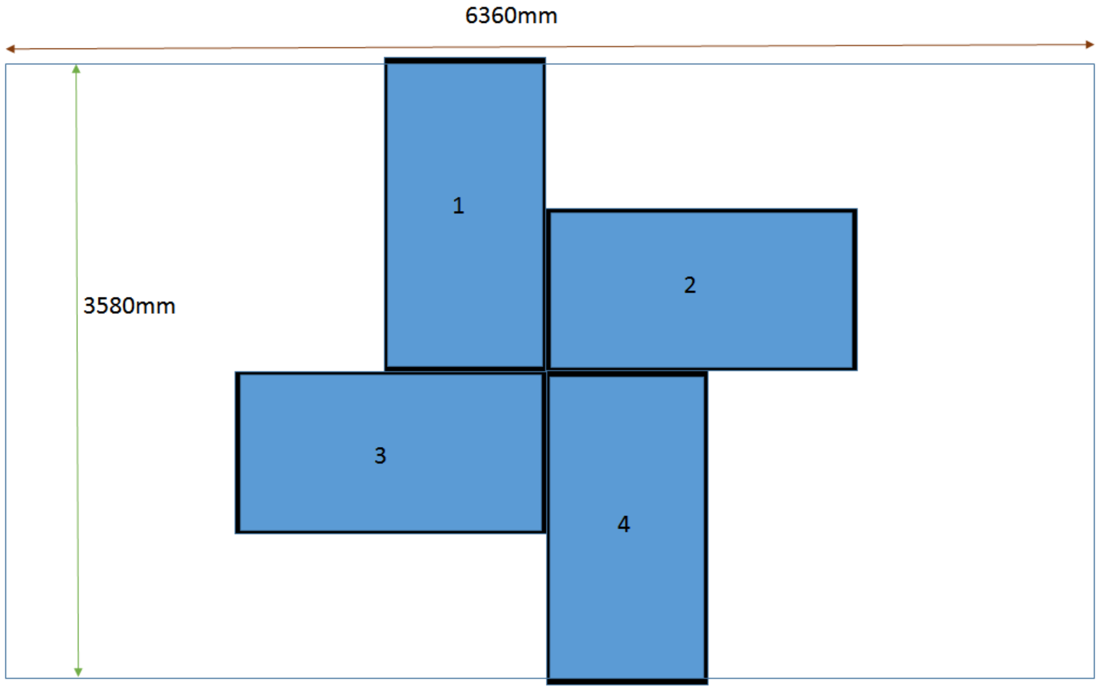
OW=200, VW=190, OH=100, VH=90

The decoders' position and commands will be:

r0c0:	r0c1:	r0c2:
X1 = 0 Y1 = 0 X2 = 3220 Y2 = 10000	X1 = 3389 Y1 = 0 X2 = 6610 Y2 = 10000	X1 = 6779 Y1 = 0 X2 = 10000 Y2 = 10000
270 degree clockwise rotate (monitor is 90 degree clockwise rotate)	270 degree clockwise rotate (monitor is 90 degree clockwise rotate)	270 degree clockwise rotate (monitor is 90 degree clockwise rotate)
#e e_vw_enable_0_0_3220_10000_1 #e e_vw_rotate_6	#e e_vw_enable_3389_0_6610_10000_1 #e e_vw_rotate_6	#e e_vw_enable_6779_0_10000_10000_1 #e e_vw_rotate_6

Mosaic Style Video Wall

Users have to calculate the position of each display area. Following is just an example for reference,



OW=1810, VW=1770, OH=1090, VH=1000 (in mm)

The decoders' position and commands will be:

1 X1 = 10000 * ((6360/ 2) - (OH - VH)/2 - VH)) / 6360 = 3356 Y1 = 0 X2 = 10000 * (((6360/ 2) - (OH - VH)/2)) / 6360) = 4929 Y2= 10000* ((3580 / 2) - ((OW - VW)/2)) / 3580 = 4944 270 degree clockwise rotate (monitor is 90 degree clockwise rotate) #e e_vw_enable_3356_0_4939_4944_2 #e e_vw_rotate_6	2 X1 = 10000 * ((6360 /2) + (OW-VW)/2) / 6360 = 5031 Y1 = 10000 * (3580 /2 - (OH-VH)/2 - VH) / 3580 = 2081 X2 = 10000 * ((6360 /2) + VW + (OW-VW)/2) / 6360 = 7814 Y2= 10000 * (3580 /2 - (OH-VH)/2) / 3580 = 4874 #e e_vw_enable_5031_2081_7814_4874_2
3 X1 = 10000 * (6360/ 2) - (OW - VW)/2 - VW) / 6360 = 2185 Y1 = 10000 * (3580/2 + (OH - VH)/2) / 3580 = 5125 X2 = 10000 * (6360/ 2) - (OW - VW)/2) / 6360 = 4968 Y2 = 10000 * (3580/2 + (OH - VH)/2 + VH) / 3580 = 7918 #e e_vw_enable_2185_5125_4968_7918_2	4 X1 = 10000 * (6360 /2 + (OH-VH)/2) / 6360 = 5070 Y1 = 10000 * (3580 /2 + (OW-VW)/2) / 3580 = 5055 X2 = 10000 * (6360 /2 + (OH-VH)/2 + VH) / 6360 = 6643 Y2 = 10000 * (3580 /2 + (OW-VW)/2 + VW) / 3580 = Ytotal/ Ytotal (where (OW-VW)/2 + VW = Ytotal/2) = 10000 270 degree clockwise rotate (monitor is 90 degree clockwise rotate) #e e_vw_enable_5070_5055_6643_10000_2 #e e_vw_rotate_6

Appendix H. KMoIP Multi-Screen Roaming

Concepts

Follow figure shows the usage of “KMoIP Multi-Screen Roaming”:

“KMoIP Multi-Screen Roaming (KMoIP Roaming)” is a special feature that allows one RX’s KMoIP function to freely connect to other TXs that connected by other RXs through moving mouse cursor across the boundary of screen and ‘roaming’ to the other screen.

In result, this feature allows user to control up to 17 computers using just one USB keyboard and mouse. K/M switch automatically and seamlessly between computers once mouse cursor crosses screen boundary.

Terms

Roaming Master: It is the RX box that you attached your keyboard and mouse for KMoIP roaming. We call it “Roaming Master”. The other involved RXs are “Roaming Slaves”.

Roaming Slave: They are other RXs that are part of the roaming setup but not “Roaming Master”.

Layout: Monitor position layout. FW uses (x,y) axis style to specify the position layout. “Roaming Master” must be (0,0).

Setup

Just configure Roaming Master’s “kmoip_roaming_layout” parameter.

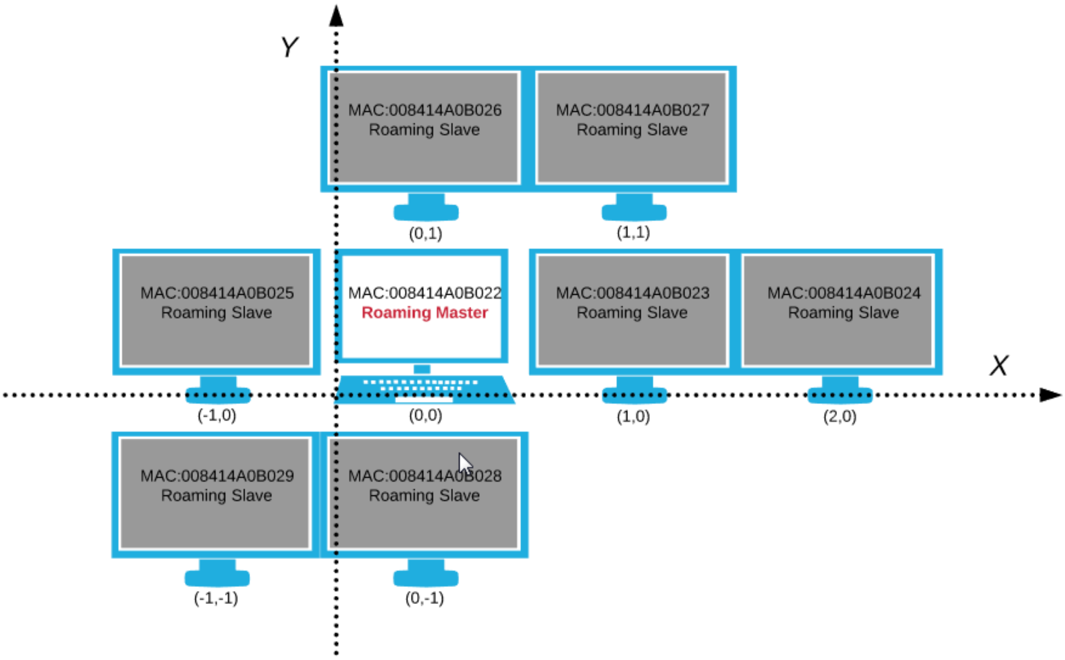
kmoip_roaming_layout:

“kmoip_roaming_layout” is a list of “MAC,X,Y” pairs. You can specify up to 16 RXs in total (no need to specify 0,0 position, so, 17 in total). Where:

MAC	A 12 characters long Ethernet MAC address of RX.
X	Horizontal layout position of the RX. Value can be -16,-15,-14,..-1,0,1,2,...,16.
Y	Vertical layout position of the RX. Value can be -16,-15,-14,..-1,0,1,2,...,16.

System reboot is required to take effect.

Example



KMoIP Multi-Screen Roaming Layout Example (kmoip_roaming_layout)

Above example's **"kmoip_roaming_layout" value will be**:

```
008414A0B026,0,1:008414A0B027,1,1:008414A0B025,-1,0:008414A0B023,1,0:008414A0B024,2,0:008414A0B029,-1,-1:008414A0B028,0,-1
```

Notes

1. (0,0) layout position is always "Roaming Master".
2. There is no need to specify roaming master (0,0) in "kmoip_roaming_layout".
3. You can specify up to 16 "Roaming Slaves". With one "Roaming Master", you can control up to 17 computers in total.
4. There is NO need to configure other RXs (Roaming Slaves). ONLY set "kmoip_roaming_layout" parameter at "Roaming Master".
5. Layout can be constructed in any style as long as the total number of roaming slaves are less then 17 (16 slaves + 1 master).
6. The "Roaming Slave" in "kmoip_roaming_layout" list can be in any order. There is no difference.
7. "Roaming Slaves" can freely switch to any video source (TX) without any extra commands. "Roaming Master" FW will monitor this change and handle it automatically.
8. The KMoIP and USBoIP function in "Roaming Slaves" are independent of

“Roaming Master”. Which means “Roaming Slave” can KMoIP/USBoIP connect to any other TXs or even disable. For example, you can use “Roaming Slave” to always attach USB storage to a PC even when K/M are roaming to another Slave.

9. The whole KMoIP and Roaming feature will not start until the “Roaming Master” connected to its encoder.

Limitations

1. “Roaming Slave/Master” doesn’t take video wall / rotate scenario into consideration. Roaming boundary and roaming target may not behave as expected.
2. Supports Windows and Linux OS. Other OSs may not work. (OS limitation)
   - Android doesn’t work.
   - MacOS works but can’t launch apps on dock.
3. Ghost key may happen if user unplug keyboard or network accidentally during operation.
4. "Roaming Slave" must be ready before "Roaming Master" starts. If not, KMoIP will not be able to roam to this slave. A "Roaming Master" e_reconnect can resolve this problem.
5. KMoIP can’t roam to secondary monitor on Windows 7.

Appendix I. List of standard keyboard codes (HID spec ch10)

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
0	00	Reserved (no event indicated)[9]	N/A	√	√	√	4/101/104
1	01	Keyboard ErrorRollOver[9]	N/A	√	√	√	4/101/104
2	02	Keyboard POSTFail[9]	N/A	√	√	√	4/101/104
3	03	Keyboard ErrorUndefined[9]	N/A	√	√	√	4/101/104
4	04	Keyboard a and A[4]	31	√	√	√	4/101/104
5	05	Keyboard b and B	50	√	√	√	4/101/104
6	06	Keyboard c and C[4]	48	√	√	√	4/101/104
7	07	Keyboard d and D	33	√	√	√	4/101/104
8	08	Keyboard e and E	19	√	√	√	4/101/104
9	09	Keyboard f and F	34	√	√	√	4/101/104
10	0A	Keyboard g and G	35	√	√	√	4/101/104
11	0B	Keyboard h and H	36	√	√	√	4/101/104
12	0C	Keyboard i and I	24	√	√	√	4/101/104
13	0D	Keyboard j and J	37	√	√	√	4/101/104
14	0E	Keyboard k and K	38	√	√	√	4/101/104
15	0F	Keyboard l and L	39	√	√	√	4/101/104
16	10	Keyboard m and M[4]	52	√	√	√	4/101/104
17	11	Keyboard n and N	51	√	√	√	4/101/104
18	12	Keyboard o and O[4]	25	√	√	√	4/101/104
19	13	Keyboard p and P[4]	26	√	√	√	4/101/104
20	14	Keyboard q and Q[4]	17	√	√	√	4/101/104

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
21	15	Keyboard r and R	20	√	√	√	4/101/104
22	16	Keyboard s and S[4]	32	√	√	√	4/101/104
23	17	Keyboard t and T	21	√	√	√	4/101/104
24	18	Keyboard u and U	23	√	√	√	4/101/104
25	19	Keyboard v and V	49	√	√	√	4/101/104
26	1A	Keyboard w and W[4]	18	√	√	√	4/101/104
27	1B	Keyboard x and X[4]	47	√	√	√	4/101/104
28	1C	Keyboard y and Y[4]	22	√	√	√	4/101/104
29	1D	Keyboard z and Z[4]	46	√	√	√	4/101/104
30	1E	Keyboard 1 and ![4]	2	√	√	√	4/101/104
31	1F	Keyboard 2 and @[4]	3	√	√	√	4/101/104
32	20	Keyboard 3 and #[4]	4	√	√	√	4/101/104
33	21	Keyboard 4 and $[4]	5	√	√	√	4/101/104
34	22	Keyboard 5 and %[4]	6	√	√	√	4/101/104
35	23	Keyboard 6 and ^[4]	7	√	√	√	4/101/104
36	24	Keyboard 7 and &[4]	8	√	√	√	4/101/104
37	25	Keyboard 8 and *[4]	9	√	√	√	4/101/104
38	26	Keyboard 9 and ([4]	10	√	√	√	4/101/104
39	27	Keyboard 0 and)[4]	11	√	√	√	4/101/104
40	28	Keyboard Return (ENTER)[5]	43	√	√	√	4/101/104
41	29	Keyboard ESCAPE	110	√	√	√	4/101/104
42	2A	Keyboard DELETE (Backspace)[13]	15	√	√	√	4/101/104
43	2B	Keyboard Tab	16	√	√	√	4/101/104
44	2C	Keyboard Spacebar	61	√	√	√	4/101/104
45	2D	Keyboard - and (underscore)[4]	12	√	√	√	4/101/104
46	2E	Keyboard = and +[4]	13	√	√	√	4/101/104
47	2F	Keyboard [and {[4]	27	√	√	√	4/101/104
48	30	Keyboard] and }[4]	28	√	√	√	4/101/104
49	31	Keyboard \ and |	29	√	√	√	4/101/104
50	32	Keyboard Non-US # and ~[2]	42	√	√	√	4/101/104
51	33	Keyboard ; and :[4]	40	√	√	√	4/101/104
52	34	Keyboard ' and "[4]	41	√	√	√	4/101/104
53	35	Keyboard Grave Accent and Tilde[4]	1	√	√	√	4/101/104
54	36	Keyboard, and <[4]	53	√	√	√	4/101/104
55	37	Keyboard . and >[4]	54	√	√	√	4/101/104
56	38	Keyboard / and ?[4]	55	√	√	√	4/101/104
57	39	Keyboard Caps Lock[11]	30	√	√	√	4/101/104
58	3A	Keyboard F1	112	√	√	√	4/101/104

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
59	3B	Keyboard F2	113	√	√	√	4/101/104
60	3C	Keyboard F3	114	√	√	√	4/101/104
61	3D	Keyboard F4	115	√	√	√	4/101/104
62	3E	Keyboard F5	116	√	√	√	4/101/104
63	3F	Keyboard F6	117	√	√	√	4/101/104
64	40	Keyboard F7	118	√	√	√	4/101/104
65	41	Keyboard F8	119	√	√	√	4/101/104
66	42	Keyboard F9	120	√	√	√	4/101/104
67	43	Keyboard F10	121	√	√	√	4/101/104
68	44	Keyboard F11	122	√	√	√	101/104
69	45	Keyboard F12	123	√	√	√	101/104
70	46	Keyboard PrintScreen[1]	124	√	√	√	101/104
71	47	Keyboard Scroll Lock[11]	125	√	√	√	4/101/104
72	48	Keyboard Pause[1]	126	√	√	√	101/104
73	49	Keyboard Insert[1]	75	√	√	√	101/104
74	4A	Keyboard Home[1]	80	√	√	√	101/104
75	4B	Keyboard PageUp[1]	85	√	√	√	101/104
76	4C	Keyboard Delete Forward[1;14]	76	√	√	√	101/104
77	4D	Keyboard End[1]	81	√	√	√	101/104
78	4E	Keyboard PageDown[1]	86	√	√	√	101/104
79	4F	Keyboard RightArrow[1]	89	√	√	√	101/104
80	50	Keyboard LeftArrow[1]	79	√	√	√	101/104
81	51	Keyboard DownArrow[1]	84	√	√	√	101/104
82	52	Keyboard UpArrow[1]	83	√	√	√	101/104
83	53	Keypad Num Lock and Clear[11]	90	√	√	√	101/104
84	54	Keypad /[1]	95	√	√	√	101/104
85	55	Keypad *	100	√	√	√	4/101/104
86	56	Keypad -	105	√	√	√	4/101/104
87	57	Keypad +	106	√	√	√	4/101/104
88	58	Keypad ENTER[5]	108	√	√	√	101/104
89	59	Keypad 1 and End	93	√	√	√	4/101/104
90	5A	Keypad 2 and Down Arrow	98	√	√	√	4/101/104
91	5B	Keypad 3 and PageDn	103	√	√	√	4/101/104
92	5C	Keypad 4 and Left Arrow	92	√	√	√	4/101/104
93	5D	Keypad 5	97	√	√	√	4/101/104
94	5E	Keypad 6 and Right Arrow	102	√	√	√	4/101/104
95	5F	Keypad 7 and Home	91	√	√	√	4/101/104
96	60	Keypad 8 and Up Arrow	96	√	√	√	4/101/104

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
97	61	Keypad 9 and PageUp	101	√	√	√	4/101/104
98	62	Keypad 0 and Insert	99	√	√	√	4/101/104
99	63	Keypad . and Delete	104	√	√	√	4/101/104
100	64	Keyboard Non-US \ and |[3,6]	45	√	√	√	4/101/104
101	65	Keyboard Application[10]	129	√		√	104
102	66	Keyboard Power[9]			√	√	
103	67	Keypad =			√		
104	68	Keyboard F13			√		
105	69	Keyboard F14			√		
106	6A	Keyboard F15			√		
107	6B	Keyboard F16					
108	6C	Keyboard F17					
109	6D	Keyboard F18					
110	6E	Keyboard F19					
111	6F	Keyboard F20					
112	70	Keyboard F21					
113	71	Keyboard F22					
114	72	Keyboard F23					
115	73	Keyboard F24					
116	74	Keyboard Execute				√	
117	75	Keyboard Help				√	
118	76	Keyboard Menu				√	
119	77	Keyboard Select				√	
120	78	Keyboard Stop				√	
121	79	Keyboard Again				√	
122	7A	Keyboard Undo				√	
123	7B	Keyboard Cut				√	
124	7C	Keyboard Copy				√	
125	7D	Keyboard Paste				√	
126	7E	Keyboard Find				√	
127	7F	Keyboard Mute				√	
128	80	Keyboard Volume Up				√	
129	81	Keyboard Volume Down				√	
130	82	Keyboard Locking Caps Lock[12]				√	
131	83	Keyboard Locking Num Lock[12]				√	
132	84	Keyboard Locking Scroll Lock[12]				√	
133	85	Keypad Comma[27]	107				
134	86	Keypad Equal Sign[29]					

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
135	87	Keyboard International1[15,28]	56				
136	88	Keyboard International2[16]					
137	89	Keyboard International3[17]					
138	8A	Keyboard International4[18]					
139	8B	Keyboard International5[19]					
140	8C	Keyboard International6[20]					
141	8D	Keyboard International7[21]					
142	8E	Keyboard International8[22]					
143	8F	Keyboard International9[22]					
144	90	Keyboard LANG1[25]					
145	91	Keyboard LANG2[26]					
146	92	Keyboard LANG3[30]					
147	93	Keyboard LANG4[31]					
148	94	Keyboard LANG5[32]					
149	95	Keyboard LANG6[8]					
150	96	Keyboard LANG7[8]					
151	97	Keyboard LANG8[8]					
152	98	Keyboard LANG9[8]					
153	99	Keyboard Alternate Erase[7]					
154	9A	Keyboard SysReq/Attention[1]					
155	9B	Keyboard Cancel					
156	9C	Keyboard Clear					
157	9D	Keyboard Prior					
158	9E	Keyboard Return					
159	9F	Keyboard Separator					
160	A0	Keyboard Out					
161	A1	Keyboard Oper					
162	A2	Keyboard Clear/Again					
163	A3	Keyboard CrSel/Props					
164	A4	Keyboard ExSel					
165-175	A5-AF	Reserved					
176	B0	Keypad 00					
177	B1	Keypad 000					
178	B2	Thousands Separator [33]					
179	B3	Decimal Separator [33]					
180	B4	Currency Unit [34]					
181	B5	Currency Sub-unit [34]					
182	B6	Keypad (					

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
183	B7	Keypad)					
184	B8	Keypad {					
185	B9	Keypad }					
186	BA	Keypad Tab					
187	BB	Keypad Backspace					
188	BC	Keypad A					
189	BD	Keypad B					
190	BE	Keypad C					
191	BF	Keypad D					
192	C0	Keypad E					
193	C1	Keypad F					
194	C2	Keypad XOR					
195	C3	Keypad ^					
196	C4	Keypad %					
197	C5	Keypad <					
198	C6	Keypad >					
199	C7	Keypad &					
200	C8	Keypad &&					
201	C9	Keypad |					
202	CA	Keypad ||					
203	CB	Keypad :					
204	CC	Keypad #					
205	CD	Keypad Space					
206	CE	Keypad @					
207	CF	Keypad !					
208	D0	Keypad Memory Store					
209	D1	Keypad Memory Recall					
210	D2	Keypad Memory Clear					
211	D3	Keypad Memory Add					
212	D4	Keypad Memory Subtract					
213	D5	Keypad Memory Multiply					
214	D6	Keypad Memory Divide					
215	D7	Keypad +/-					
216	D8	Keypad Clear					
217	D9	Keypad Clear Entry					
218	DA	Keypad Binary					
219	DB	Keypad Octal					
220	DC	Keypad Decimal					

Usage ID (Dec)	Usage ID (Hex)	Usage Name	Ref: Typical AT-101 Position	PC-AT	Mac	UNIX	Boot
221	DD	Keypad Hexadecimal					
222-223	DE-DF	Reserved					
224	E0	Keyboard LeftControl	58	√	√	√	4/101/104
225	E1	Keyboard LeftShift	44	√	√	√	4/101/104
226	E2	Keyboard LeftAlt	60	√	√	√	4/101/104
227	E3	Keyboard Left GUI[10;23]	127	√	√	√	104
228	E4	Keyboard RightControl	64	√	√	√	101/104
229	E5	Keyboard RightShift	57	√	√	√	4/101/104
230	E6	Keyboard RightAlt	62	√	√	√	101/104
231	E7	Keyboard Right GUI[10;24]	128	√	√	√	104
232-65535	E8-FFFF	Reserved					

Appendix J. Fast Switching Mode

Concepts

For fast switching be enabled, there are several settings need be changed:
1. Setup Encoder's EDID as fast switching mode.
2. Setup Decoder's output resolution as auto (EDID preferred resolution output).
3. Setup Decoder's HDCP policy as follow output.

Setup

On Encoder apply below API

```
e e_edid_fast_switch_mode::y
```

On Decoder apply below APIs:

```
astparam s v_output_timing_convert 82000000; astparam s hdcp_always_on y; astparam s hdcp_always_on_22 n; astparam save; reboot
```

Note: 8200000 output resolution is EDID based, if user would like to use other resolution as output, it can be changed to any other resolution, but it should not be passthrough mode.